CVMA

MIVP Dep't

# Veterinary Microbiology-II (Vetm3085)

## Specific topic: **Virology**

**For DVM Year III Students**

**By:**

**Debebe A.**

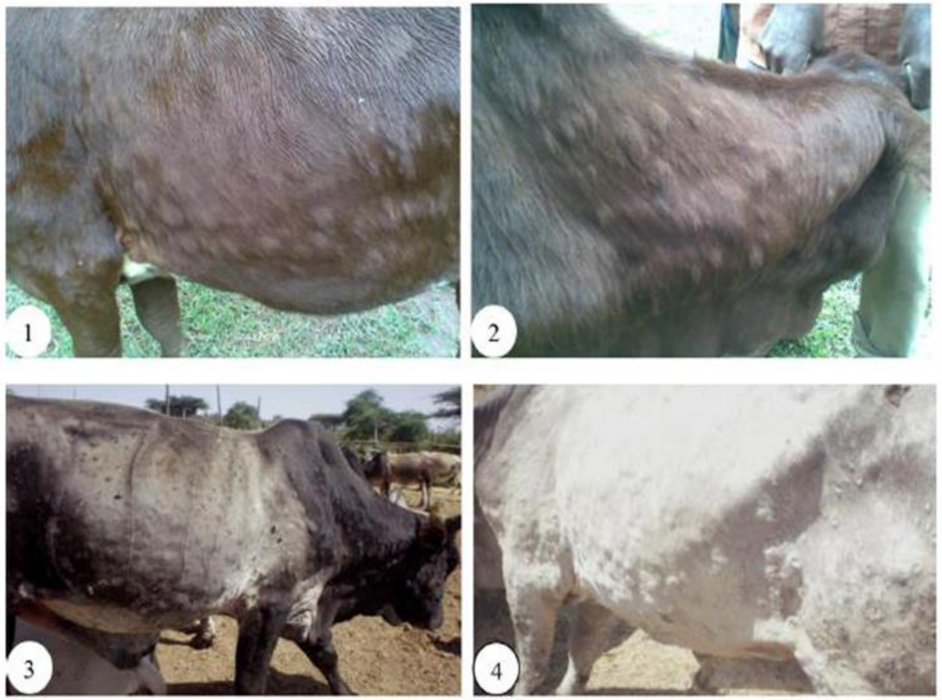
# Outline

- **II. Systematic Virology**
  - 1. **DNA Virus Family**
    - 1.1. Poxviridae
    - 1.2. Adenoviridae
    - 1.3. Herpesviridae
    - 1.4. Parvoviridae
    - 1.5. Papovaviridae
    - 1.6. Asfarviridae

- **2. RNA Virus Family**
  - 2.1. Rhabdoviridae
  - 2.2. Picornaviridae
  - 2.3. Retroviridae
  - 2.4. Orthomyxoviridae
  - 2.5. Paramyxoviridae
  - 2.6. Bunyaviridae
  - 2.7. Flaviviridae
  - 2.8. Togaviridae
  - 2.9. Reoviridae
  - 2.10. Coronaviridae
  - 2.11. Birnaviridae
- **3. Prions**
  - 3.1. Disease in animals and human beings caused by prions

# Poxviridae

- The family *Poxviridae* includes numerous viruses of veterinary and/or medical importance.
- Poxviruses are large DNA viruses that are capable of infecting both invertebrates and vertebrates.
- Poxvirus diseases occur in most animal species, and are of considerable economic importance in some regions of the world.
- A characteristic of many of these viruses is their common ability to induce characteristic "pox" (pockmark) lesions in the skin of affected animals.
- Other feature???



# Properties of Poxviruses

## Classification

- The family *Poxviridae* is subdivided into two subfamilies:
  - *Chordopoxvirinae* (poxviruses of vertebrates) and
  - *Entomopoxvirinae* (poxviruses of insects).
- The subfamily *Chordopoxvirinae* is subdivided into eight genera.
  - Each of the genera, includes species that cause diseases in domestic or lab animals.
- Because of the large size and distinctive structure of poxvirus virions, negative-stain electron microscopic examination of lesion material is used for diagnosis, but it does not allow specific verification of virus species or variants.
- Molecular methods identify additional pathogenic poxvirus species; E.g. *Entomopoxvirinae* (proliferative gill disease in farmed salmon).

# Genera in *Chordopoxvirinae*: *host range & geographic distribution*

| Genus | Virus | Hosts | Geographic distribution |
|---|---|---|---|
| *Orthopoxvirus* | Viriola (smallpox) virus | Human | ? |
| | Viccina virus | Human, cattle, buffaloes, swine, rabbit | Worldwide |
| | Cowpox virus | Rodents, domestic cat, cattle, humans, elephants, large felids | Europe and Asia |
| | Camelpox virus | Camel | Africa, Asia |
| *Capripoxvirus* | Sheeppox virus | Sheep, Goats | Africa, Asia |
| | Goatpox virus | Sheep, Goats | Africa, Asia |
| | Lumpy skin disease virus | Cattle, Cape buffalo | Africa, (+Middle East & Europe) |
| *Avipoxvirus* | Fowlpox virus, Turkeypox virus | Chicken, turkey, other birds | Worldwide |
| *Suipoxvirus* | Swinepox virus | Swine | Worldwide |
| *Parapoxvirus* | Orf virus | Shoats, humans | Worldwide |

# Cont…

## ❑Virion Properties

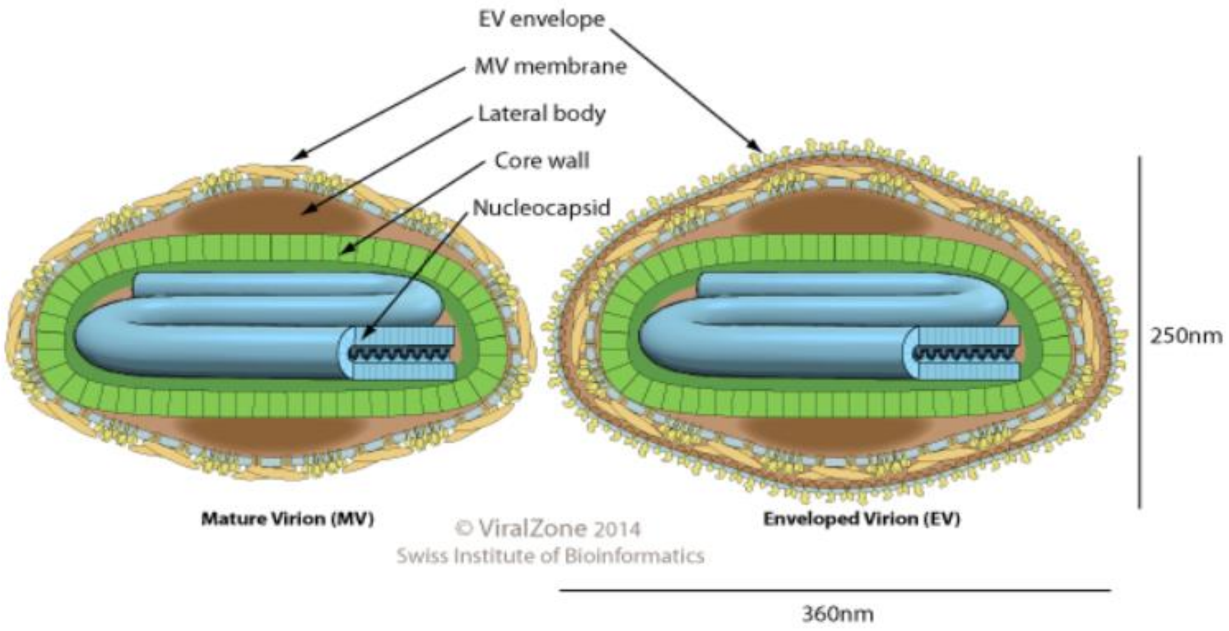
- Most poxvirus virions are pleomorphic, typically brick shaped (220–450 nm x 140–260 nm) wide with an irregular surface of projecting tubular or globular structures,
- Those of the genus *Parapoxvirus* are ovoid (250–300 nm long and 160–190 nm in diameter) with a regular surface and are covered with long thread-like surface tubules, which appear to be arranged in crisscross fashion, resembling a ball of yarn.
- The virion outer layer encloses a dumbbell-shaped core and two lateral bodies.
- The core contains the viral DNA, together with several proteins.
- There is no isometric nucleocapsid conforming to either icosahedral or helical symmetry; hence poxviruses are said to have a “complex” structure.
- Virions that are released from cells by budding, rather than by cellular disruption, have an extra envelope that contains cellular lipids and several virus-encoded proteins.

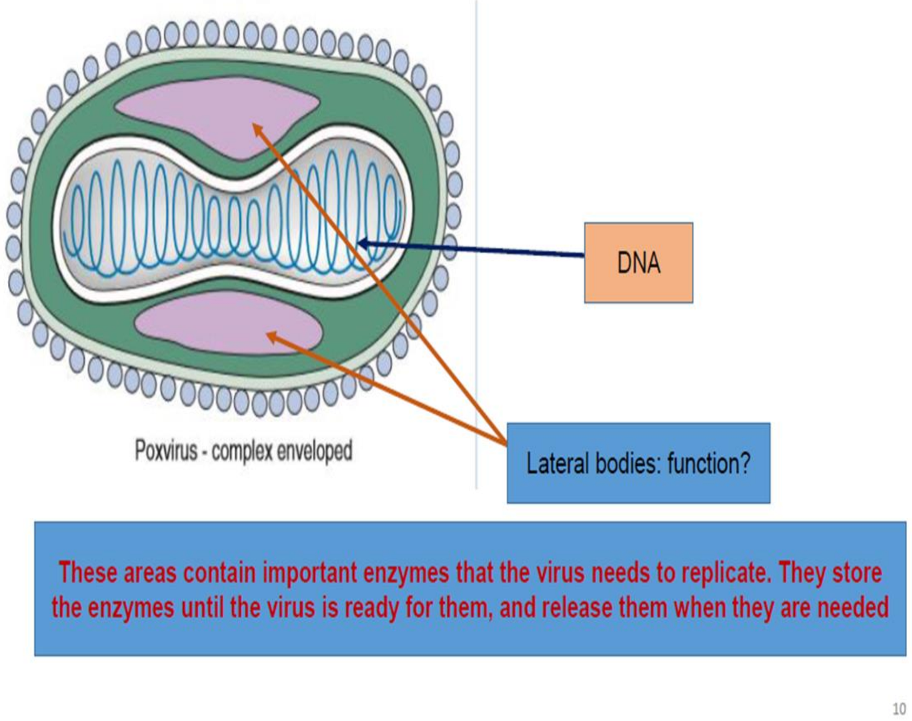
## Cont…

- The genome consists of a single molecule of linear dsDNA varying in size from 130 kbp (parapoxviruses), to 280 kbp (fowlpox virus), up to 375 kbp (entomopoxviruses).
- Poxvirus genomes have cross-links that join the two DNA strands at both ends; the ends of each DNA strand have long inverted tandemly repeated nucleotide sequences that form single-stranded loops.
- Poxviruses have more than 200 genes in their genomes, and 100 of these encode proteins that are contained in virions.
- Many of the viral proteins with known functions are enzymes involved in nucleic acid synthesis and virion structural components.
    - E.g. DNA polymerase, DNA ligase, RNA polymerase, enzymes involved in capping and polyadenylation of mRNAs, and thymidine kinase.

# Poxviridae

**VIRION**

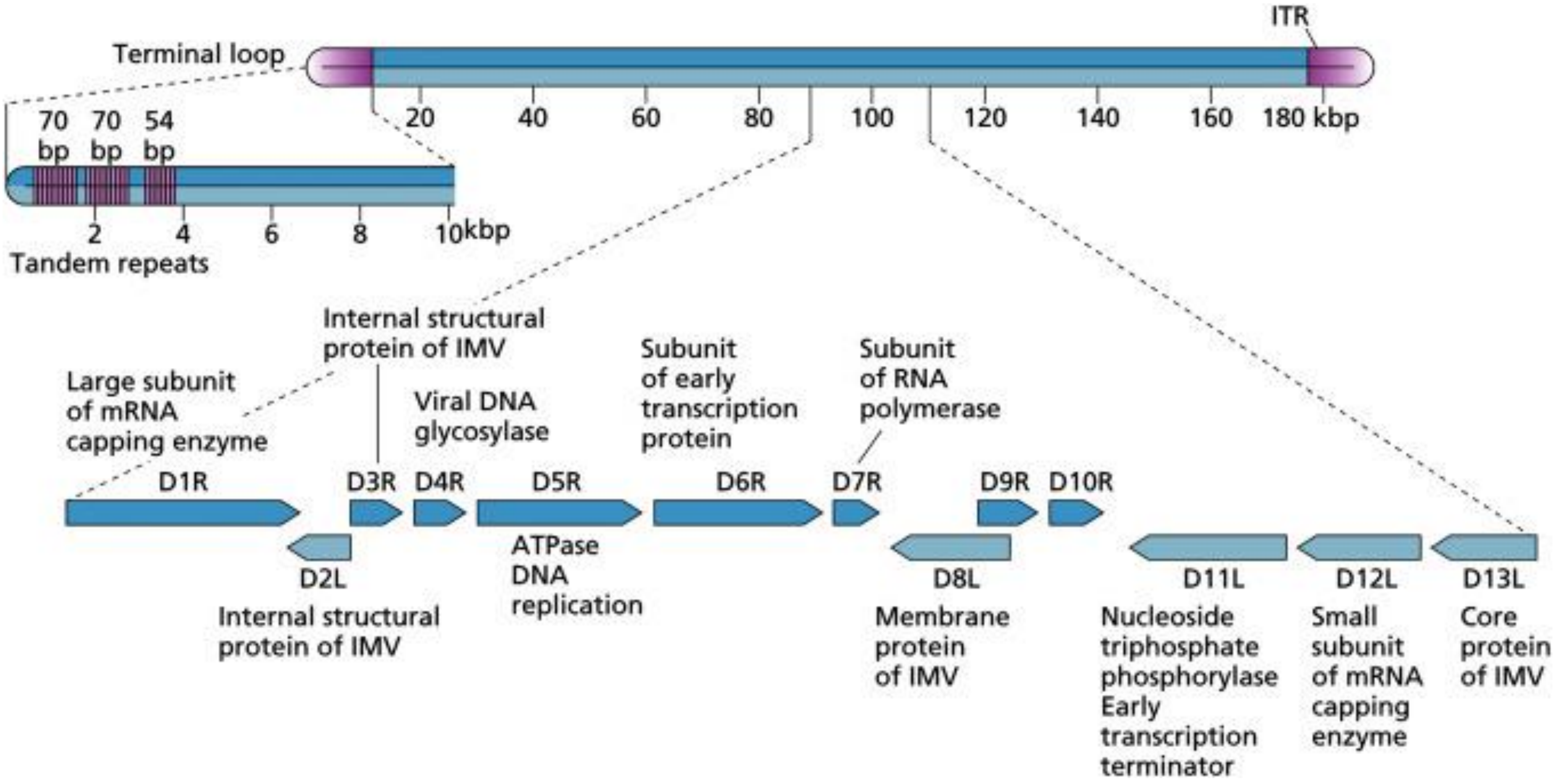



Enveloped, brick-shaped or ovoid virion, 220-450 nm long and 140-260 nm wide. The surface membrane displays surface tubules or surface filaments. Two distinct infectious virus particles exists: the intracellular mature virus (IMV) and the extracellular enveloped virus (EEV).
Vaccinia virion 3D tour

**GENOME**

Linear, dsDNA genome of 130-375kb. The linear genome is flanked by inverted terminal repeat (ITR) sequences which are covalently-closed at their extremities.



ITR
Terminal loop
20
40
60
80
100
120
140
160
180 kbp
70 bp
70 bp
54 bp
2
4
6
8
10 kbp
Tandem repeats
Internal structural protein of IMV
Large subunit of mRNA capping enzyme
Viral DNA glycosylase
Subunit of early transcription protein
Subunit of RNA polymerase
D1R
D3R
D4R
D5R
D6R
D7R
D9R
D10R
D2L
ATPase DNA replication
D8L
D11L
D12L
D13L
Internal structural protein of IMV
Membrane protein of IMV
Nucleoside triphosphate phosphorylase Early transcription terminator
Small subunit of mRNA capping enzyme
Core protein of IMV

# Cont…

- The genomes also include a remarkable number of genes encoding proteins that specifically counteract host adaptive and innate immune responses.
- Poxviruses are transmitted between animals by several routes:
    - by introduction of virus into skin abrasions, or
    - directly or indirectly from a contaminated environment.
- Several poxviruses, including sheeppox, swinepox, fowlpox, and myxoma virus, are transmitted mechanically by biting arthropods.
- Poxviruses are generally indigenous to specific host niches, but, in many cases, they are not species-specific.
- Poxviruses are resistant in the environment under ambient temperatures, but can survive for many years in dried scabs or other virus-laden material.

# Cont…

## ❑Virus Replication

- Replication of poxviruses occurs predominantly in the cytoplasm.
- To achieve this, they have evolved to encode the enzymes required for transcription and replication of the viral genome, several of which must be carried in the virion itself.
- Replication begins after fusion of the extracellular enveloped virion with the PM, or after endocytosis, and the virus core is then released into the cytoplasm where it "uncoats".
- Transcription is characterized by a cascade in which intermediate gene transcription factors are encoded by early genes, whereas late transcription factors are encoded by intermediate genes.
- Host macromolecular synthesis is inhibited with the production of viral proteins.
- Poxvirus DNA replication involves the synthesis of long concatameric intermediates, which are subsequently cut into unit-length genomes that are ultimately covalently linked.

# Cont...

FOWL POX

- Fowl pox (FP) characterized by cutaneous lesions on the feather-less skin areas.
- In most outbreaks, the cutaneous form is prevailing. The lesions vary according to the stage of development: papules (small ones on the comb, wattles, and around the beak), vesicles, pustules or crusts that can occur.
- The nodules become yellowish and progress to a thick dark scab. The lesions are usually in the region of the head & sometimes on legs and feet and around the cloaca.
- Nasal discharge & excessive lacrimation are common.

- The diphtheritic form causes lesions on mucous coats of the upper alimentary and respiratory tracts. It is probably caused by droplet infection and involves infection of the mucous membranes of the mouth, pharynx, larynx, and sometimes the trachea. As the lesions coalesce, it results in a necrotic pseudomembrane, which can cause death by asphyxiation. The prognosis for this form of fowlpox is poor.

# Diagnosis

- **LSD:**
  - Clinical signs: fever, nodular lesions in the skin that leads to necrosis.
  - Morbidity: up to 100%
  - Mortality :1-2%
- The economic importance of the disease relates to the prolonged convalescence and, in this respect, lumpy skin disease is similar to foot-and-mouth disease.
- **Samples:** biopsy or at post-mortem from skin nodules, lung lesions or lymph nodes, Buffy coat from blood
- **1. LSD Ag identification:**
  - 1.1). Virus isolation on cell-culture: LSD virus will grow in tissue culture of bovine, ovine or caprine origin, although primary or secondary culture of bovine dermis cells or lamb testis (LT) cell are most susceptible.
  - 1.2). Electron microscopy of biopsy samples
  - 1.3). Fluorescent antibody tests. It needs grow the virus on tissue culture slides or/and cover slips
  - 1.4). Agar gel immunodiffusion

## Cont...

- 1.5). Enzyme-linked immunosorbent assay: needs antisera (Ab) and conjugate and spectrophotometer (ELISA reader).
- 1.6). PCR : needs extraction of DNA, primers (both Forward and reverse), thermo cycler, and gel documentation apparatus.

- **2. Serological tests:**
  - Test sera are needed and tested against known LSD antigen.
    - 2.1. Neutralization test
    - 2.2. Indirect immunofluorescence test
    - 2.3. ELISA
    - 2.4 Western or immune-blotting

**►Diagnostic approaches for other poxvirus caused diseases???**

# Herpesviridae

- Herpesviruses have been found in insects, fish, reptiles, amphibians, and mollusks as well as in virtually every species of bird and mammal that has been investigated.
- It is likely that every vertebrate species is infected with several Herpesvirus species.
- At least one major disease of each domestic animal species, except sheep, is caused by a Herpesvirus.
- Are adapted to their individual hosts, probably as a consequence of prolonged co-evolution.
- Thus, with some exceptions (subfamily *Alphaherpes virinae)*, Herpesvirus infections typically produce severe disease only in neonates, fetuses, immunocompromised individuals, or in alternate host species (so-called species-jumping).
- The virions are easily inactivated and do not survive well outside the body.
- In general, transmission requires close contact, particularly mucosal contact (e.g., coitus, licking and nuzzling, as between mother and offspring or between neonates).
- In large, closely confined populations, such as found in cattle feedlots, modern swine farrowing units, animal shelters, catteries, or broiler facilities, sneezing and short distance droplet spread are major modes of transmission.

## Cont…

- However, moist, cool environmental conditions provide opportunity for transmission over longer distances.
- Similarly, during active herpesvirus outbreaks in fish, virus shed into the water may spread rapidly between individuals in densely stocked ponds.
- In addition, vertical transmission from adults to progeny may be the major mode by which herpesviruses are maintained in wild and captive fish populations.
- An important aspect of Herpesvirus pathogenesis is latency.
- Latency is defined as persistent life-long infection of a host with restricted but recurrent virus replication.
- Recurrent virus replication can lead to shedding, transmission, and the maintenance of detectable antiviral immune responses.
- Therefore, latent infections in clinically normal hosts provide a potentially undiagnosed reservoir for virus transmission.

# Properties of Herpesviruses

## Classification

- The classification of herpesviruses is complex.
- All herpesviruses share a common morphology and have genomes of linear, dsDNA .
- The herpesviruses were recently assigned to the new order *Herpesviruses*, with three distinct families:
  - the *Herpesviridae* that includes herpesviruses of mammals, birds, and reptiles;
  - the *Alloherpesviridae* that includes the *herpesviruses* of fish and frogs, and
  - the *Malacoherpesviridae* that contains a virus of oysters (bivalve).
- A substantial number of viruses have not yet been assigned to specific genera.
- Antigenic relationships among the herpesviruses are complex; there are some shared antigens within the order, but different species have distinct envelope glycoproteins.

# Cont…

- Besides, herpesviruses are classified into three groups based upon of tissue tropism, pathogenicity and behavior:

**α herpesviruses**

- Fast replicating
- Variable host range
- Typically destroy host cell
- Latency established in sensory ganglia, e.g. Herpes Simplex virus-1 and 2 (HSV-1, HSV-2), Varicella-Zoster virus (VZV)

**β herpesviruses**

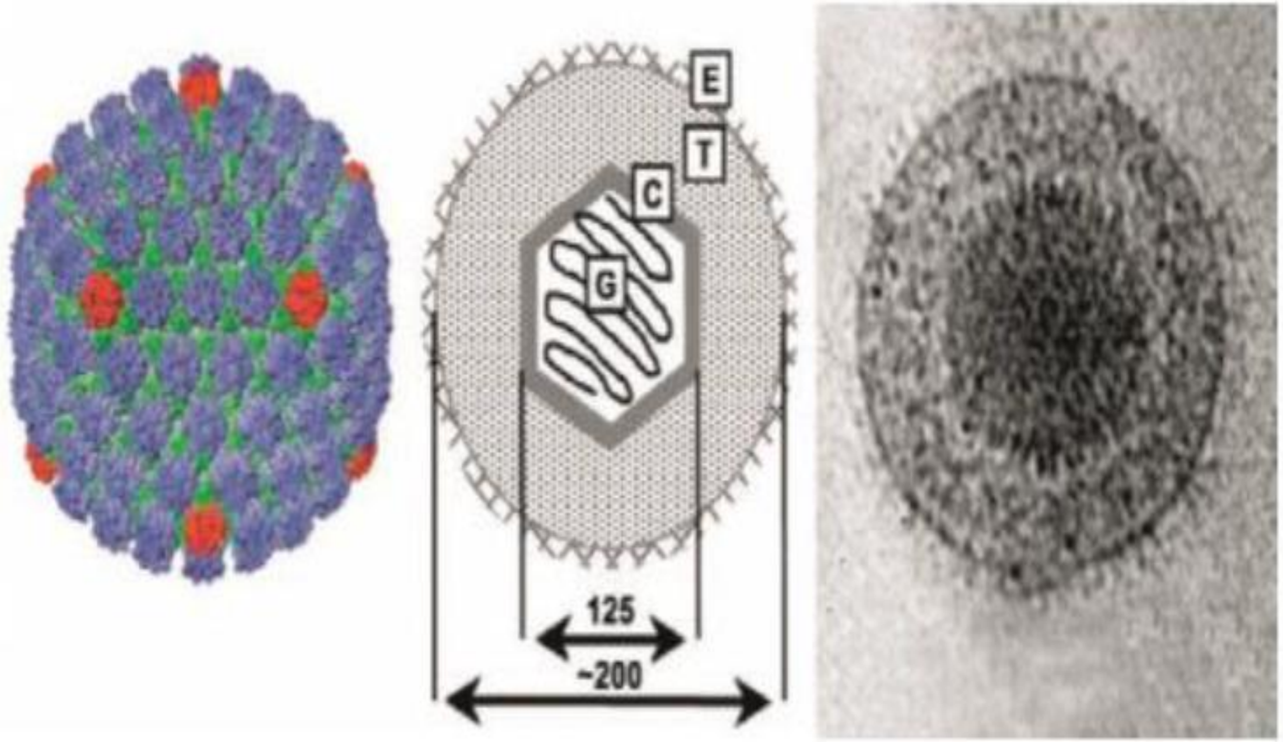
- Slowly replicating
- Restricted host range
- Infected cells enlarge (*cytomegalia*)
- Latency established in secretory glands, lymphoreticular cells, kidneys, e.g. Cytomegalovirus (CMV), Human Herpesvirus- 6 and 7 (HHV-6 , HHV-7)

**γ herpesviruses**

- Replicate poorly
- Highly restricted host range
- Latency established in lymphoid tissue (T-cell or B-cell specific), e.g. Epstein-Barr Virus (EBV), a B-cell transforming virus; Human Herpesvirus-8 (HHV-8, KSHV)

# Cont....

## Herpesvirus Morphology




Herpesvirus morphology. (Left) Reconstruction of a human herpesvirus 1 (HHV-1) capsid generated from cryo-electron microscope images, viewed along the 2-fold axis. The hexons are shown in blue, the pentons in red, and the triplexes in green. (Courtesy of W. Chiu and H. Zhou). (Center) Schematic representation of a virion with diameters in nm. (G) genome, (C) capsid, (T) tegument, (E) envelope. (Right) Cryo-electron microscope image of a HHV-1 virion. (Reproduced from Rixon (1993) with permission from Elsevier). *[From* Virus Taxonomy: Eighth Report of the International Committee on Taxonomy of Viruses *(C. M. Fauquet, M. A. Mayo, J. Maniloff, U. Desselberger, L. A. Ball, eds), p. 193. Copyright © Elsevier (2005), with permission.]*

# Animal diseases caused by Herpesviruses

| Virus | Genus | Comments |
|---|---|---|
| Bovine herpesvirus 1 | Varicellovirus | Causes respiratory (infectious bovine rhinotracheitis) and genital (infectious pustular vulvovaginitis, balanoposthitis) infections. Occurs worldwide |
| Bovine herpesvirus 2 | Simplexvirus | Causes ulcerative mammillitis in temperate regions and pseudolumpy-skin disease in tropical and subtropical regions |
| Gallid herpesvirus 2 | Marek's disease like viruses | Causes Marek's disease, A lymphoproliferative condition in 12 to 24 week old chickens. Occurs worldwide |
| Ovine herpesvirus 2 | Rhadinovirus | Causes subclinical infection in sheep and goats worldwide. Causes malignant catarrhal fever in cattle and in some wild ruminants |
| Equine herpesvirus 1 | Varicellovirus | Causes abortion, respiratory disease, neonatal infection and neurological disease. Occurs worldwide |
| Porcine herpesvirus 1(Aujeszky's disease virus) | Varicellovirus | Causes Aujeszky's disease (pseudorabies) primarily in pigs. Encephalitis, pneumonia and abortion are features of the disease. In many species other than pigs, pseudorabies manifests as a neurological disease with marked pruritis. Occurs worldwide |

# Cont...

## Gallid Herpesvirus 2 (Marek's Disease Virus)-cont'd

- The causative agent was identified as a herpesvirus in 1967.
- Vaccination was introduced in 1970.
- Marek's disease was the most common lymphoproliferative disease of chickens, causing substantial economic losses worldwide.
- **Transmission**: Birds get infected by inhalation of infected dust from the poultry houses, and
- Following a complex life cycle, the virus is shed from the feather follicle of infected birds.
- MD occurs at 3–4 weeks of age or older and is most common between 12 and 30 weeks of age.
- MD is associated with several distinct pathological syndromes, of which the lymphoproliferative syndromes are the most frequent and are of most practical significance.

# Cont…

## 5. Marek's Disease

*(Herpesvirus)*

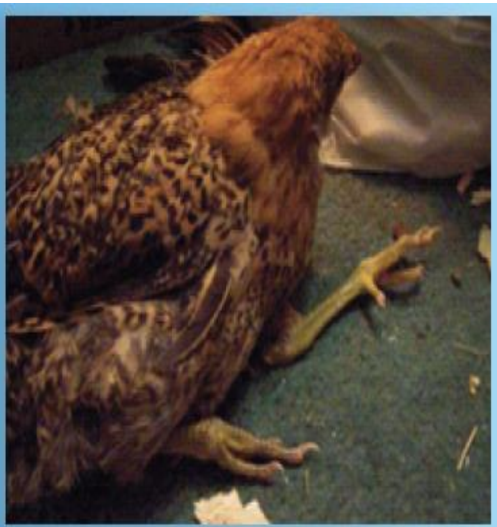
Neurological form

( progressive paralysis):

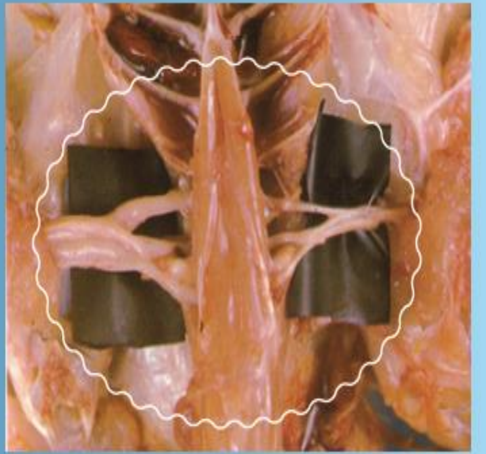
Paralysis (loss of muscle function) of wings, characteristic dropping of limb.

Twisted neck (torticollis)

Lameness.



Brachial plexus (nerve) is two or three times the normal thickness, swelling caused by fluid (oedema).



# Cont...

## Diagnosis of Marek's disease

- Diagnosis of MD should be based on a combination of MDV isolation or detection of the genome by polymerase chain reaction (PCR) and clinical disease. **Why?**
- 1. Identification of the agent,
- a) Virus isolation
- Samples needed  buffy coat cells from heparinised blood samples, or suspensions of lymphoma cells or spleen cells (or feather tips).
- Cells used monolayer cultures of chicken kidney cells or duck embryo fibroblasts (chicken embryo fibroblasts (CEF)).
- Serotype 2 and 3 viruses are more easily isolated in CEF than in chicken kidney cells.
- Culture medium is replaced at 2-day intervals. Areas of cytopathic effects, termed plaques, appear within 3–5 days and can be enumerated at about 7–10 days.

# Cont...

- b) Antigen detection: A variation of the AGID test
- c) Polymerase chain reaction: Genomes of all three serotypes have been completely sequenced.
-
- 2. **Serological tests**. The presence of antibodies to MDV in non-vaccinated chickens from about 4 weeks of age is an indication of infection. Before that age, such antibodies may represent maternal transmission of antibody via the yolk and are not evidence of active infection.
- a) Agar gel immunodiffusion
- b) ELISA

- **Control**: vaccination (shortly post hatch or in ovo); as it needs 7-14 days for full protection

# Adenoviridae

- Adenoviruses are DNA viruses first isolated from adenoidal tissue in 1953.
- This family consists of double-stranded DNA viruses with an icosahedral nucleocapsid.
- They have been recovered from many mammalian and avian species.
- Many are found in the respiratory tract and infections are often persistent.
- Only a small number cause significant veterinary diseases.
- It is likely that all vertebrate species, from fish to mammals, have their own unique adenovirus or adenoviruses with which they have co-evolved.
- Most of these viruses produce subclinical infections in their respective hosts, with occasional upper respiratory disease, but canine and avian adenoviruses are especially associated with clinically important disease syndromes.
- Since their discovery, adenoviruses have been at the core of significant basic discoveries concerning virus structure, eukaryotic gene expression and organization, RNA splicing, and apoptosis.

## Cont'…

- Adenoviruses are frequently used as experimental vectors for gene therapy and cancer therapy, and have been used as vectors for recombinant vaccines.
- Specifically, some of the adenoviruses of humans, cattle, and chickens cause tumors when inoculated into newborn lab animals and have been used in experimental oncogenesis studies; however, none has been proven to cause tumors in their respective natural hosts.

## Vector in Gene Therapy

- Adenoviral genomes have been developed into vectors in experimental therapy since adenoviruses readily infect human and other mammalian cells.
- Vector genomes carry deletions in the E1 and E3 regions; the gaps in the genome are used to take up foreign genes, e.g., the gene for the cystic fibrosis transmembrane conductance regulator (CFTR).
- Deletions in E1 minimize the potential of these vector genomes to elicit an infection cycle in human cells.
- The first clinical applications in patients suffering from the genetic disease cystic fibrosis have been reported but problems with adenovirus toxicity remain.

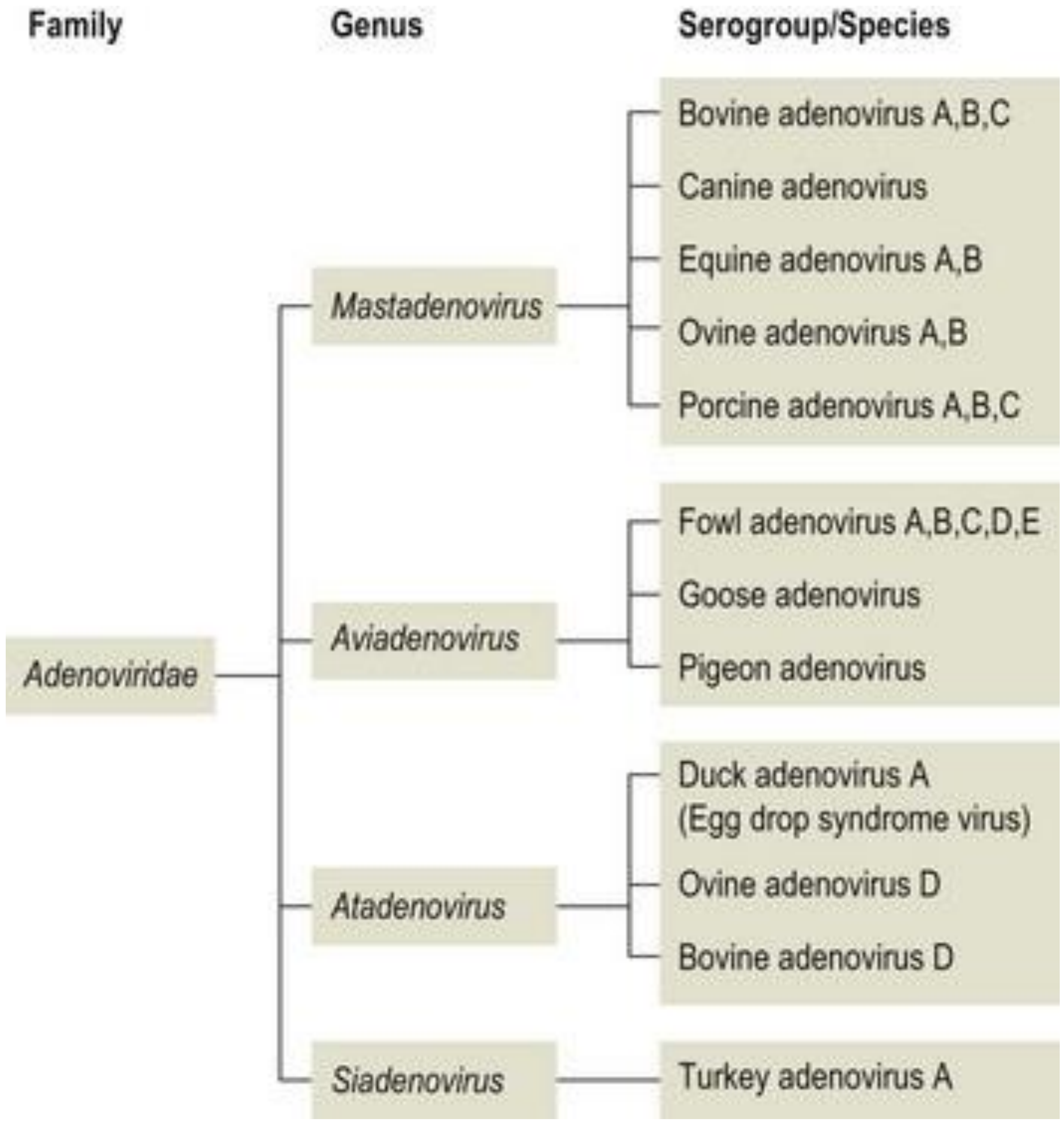
# Properties of Adenoviruses

## Classification

- The family *Adenoviridae* currently comprises four serologically distinct genera:

1) the genus *Mastadenovirus*, comprising viruses that infect only mammalian species;
2) the genus *Aviadenovirus, comprising viruses that infect only* birds;
3) the genus *Atadenovirus* that includes viruses that infect a broad host range, including snakes, lizards, ducks, geese, chickens, possums, and ruminants;
4) the genus *Siadenovirus*, which includes frog adenovirus 1 and turkey adenovirus 3, plus several recently described viruses of raptors, budgerigars, and tortoises.

- Although all adenoviruses have a similar morphology, the genomic organization differs between viruses in the various genera.
- Mastadenoviruses contain the unique proteins V and IX; protein V is involved in transport of viral DNA to the cell nucleus and protein IX is a transcriptional activator.
- Genes encoding proteins V and IX are absent in aviadenoviruses and their genomes are 20– 45% larger than those of mastadenoviruses.
- Atadenoviruses encode a unique structural protein, p32K, and apparently lack the immunomodulatory proteins that occur in the E3 region of mastadenoviruses.
- The genomic structure of siadenoviruses is also unique in that genes encoding proteins V and IX are absent as well as the genes encoding early regions E1, E3, and E4 of mastadenoviruses.



Family
Genus
Serogroup/Species
Adenoviridae
Mastadenovirus
Bovine adenovirus A,B,C
Canine adenovirus
Equine adenovirus A,B
Ovine adenovirus A,B
Porcine adenovirus A,B,C
Aviadenovirus
Fowl adenovirus A,B,C,D,E
Goose adenovirus
Pigeon adenovirus
Atadenovirus
Duck adenovirus A
(Egg drop syndrome virus)
Ovine adenovirus D
Bovine adenovirus D
Siadenovirus
Turkey adenovirus A

# Virion Properties

- Adenovirus virions are non-enveloped, precisely hexagonal in outline, with icosahedral symmetry, 70–90 nm in diameter.
- Virions are composed of 252 capsomers: 240 hexons that occupy the faces and edges of the 20 equilateral triangular facets of the icosahedron and 12 pentons (vertex capsomers) that occupy the vertices.
- The hexons consist of two distinct parts: a pseudohexagonal base with a hollow center, and a triangular top that includes three distinct "towers."
- From each penton projects a penton fiber 9–77.5 nm in length, with a terminal knob.
- The genome of a single linear molecule of dsDNA, 26–45 kbp in size, with inverted terminal repeats.

# Adenovirus structure and genomic organization

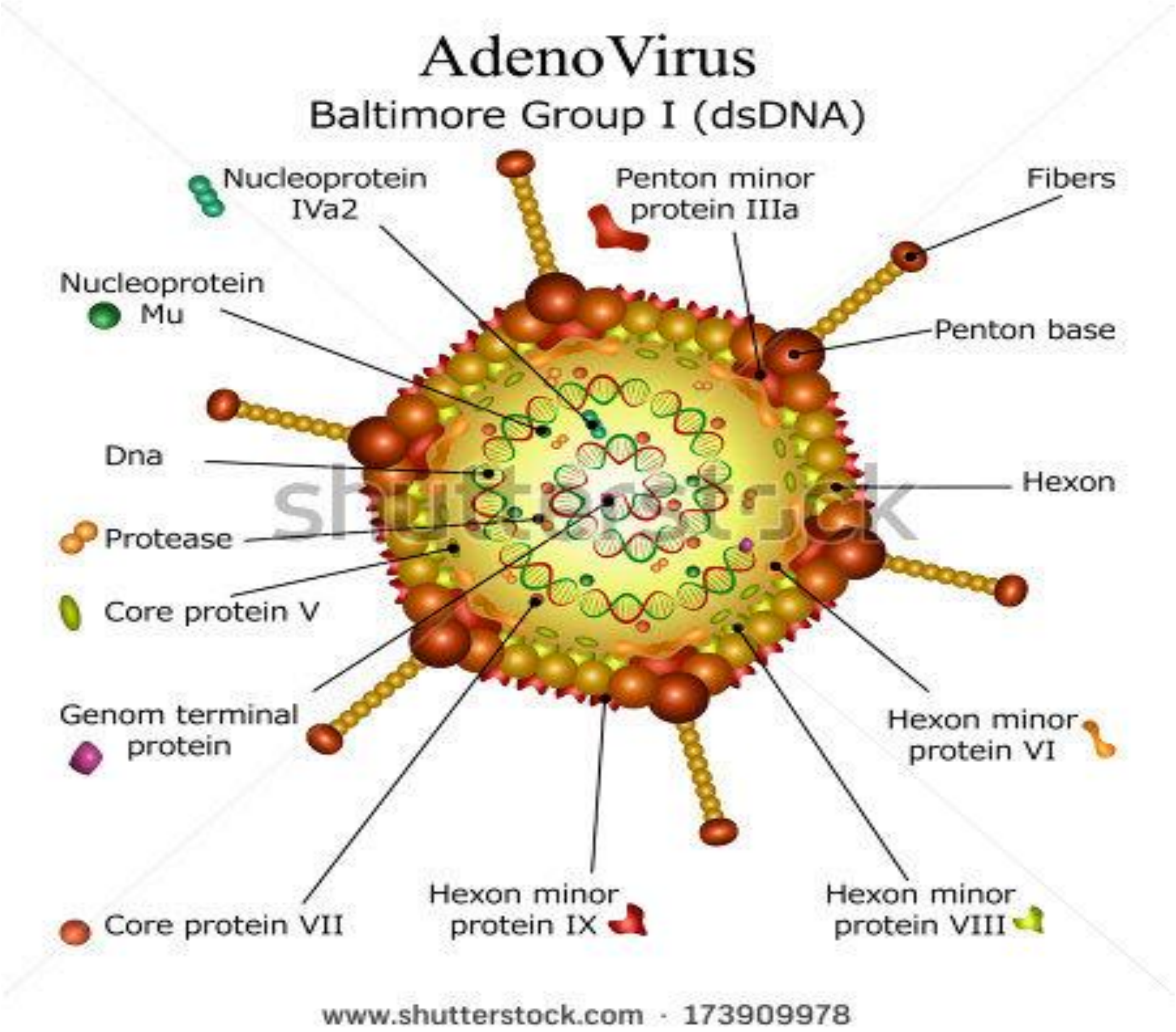


**Table 62.1.** Diseases of Domestic Animals Caused by Adenoviruses

| Virus | Type of Disease |
|---|---|
| *Mastadenovirus* | |
| Bovine adenoviruses Types 1–10 | Conjunctivitis, pneumonia, diarrhea, polyarthritis |
| Canine adenovirus Type 1 (CAV-1; infectious canine hepatitis) | Hemorrhagic and hepatic |
| Canine adenovirus Type 2 | Respiratory |
| Equine adenovirus Types 1–2 | Pneumonia |
| Ovine adenoviruses Types 1–6 | Respiratory and enteric |
| Porcine adenoviruses Types 1–4 | Diarrhea or meningoencephalitis, or both |
| *Aviadenovirus* | |
| Chicken adenoviruses Types 1–12 | Respiratory disease, enteric disease, egg-drop syndrome, aplastic anemia, atrophy of the bursa of Fabricius |
| Turkey adenoviruses Types 1–4 | Respiratory disease, enteritis, marble spleen disease |

# Adenovirus

Cornell University
College of Veterinary Medicine
Animal Health Diagnostic Center

## BASICS

Adenoviruses are double-stranded DNA viruses that infect all groups of vertebrates, including mammals, birds, reptiles, and fish. Most adenoviruses are species specific, but some types can infect a variety of hosts.

Adenoviruses cause a wide range of illnesses in animals and humans, with the respiratory and gastrointestinal systems most commonly affected. Adenovirus infections may be asymptomatic in hosts.

**CLINICAL SIGNS** of adenovirus infection vary according to the virus type and animal species infected. Red foxes infected with canine adenovirus-1 show signs of **ENCEPHALITIS** such as seizures, paralysis, coma, and death.

Skunk adenovirus affects the **RESPIRATORY SYSTEM** of a variety of animals, including porcupines, leading to nasal and ocular discharge and pneumonia.

In deer, adenovirus infection causes a **HEMORRHAGIC DISEASE** often resulting in rapid death.

**TRANSMISSION** of adenoviruses occurs through direct contact between infected animals or indirect contact via nasal secretions, urine, and feces. Spread may also occur through airborne routes, contaminated water, and contaminated equipment. Some avian adenoviruses can be transmitted from mother to chick through the egg.

Adenovirus infection is **DIAGNOSED** at necropsy with gross findings, histopathology with adenovirus inclusion bodies, and virus detection using PCR or virus isolation from infected tissues.

There is **NO TREATMENT** for adenoviral disease other than supportive care.

Vaccines to **PREVENT** specific adenovirus infections in domestic dogs and poultry are available. There are no vaccines to prevent adenovirus infections in wildlife species.


## DETAILS

**CANINE ADENOVIRUS-1** (CAdV-1) causes infectious hepatitis in domestic dogs, an acute illness affecting the liver and respiratory tract that often leads to sudden death. CAdV-1 can also infect wild carnivores including foxes, wolves, coyotes, bears, skunks, and raccoons. Infection in red foxes may result in liver damage and vasculitis in the central nervous system. **CLINICAL SIGNS** include anorexia, jaundice, nasal discharge, diarrhea, seizures, paralysis, coma, and death. Some infections may remain asymptomatic. Fatal infections are more common in juvenile foxes. Gray foxes may experience diarrhea, ocular discharge, and hepatitis.

**SKUNK ADENOVIRUS** (SkAdV-1) is unusual in that it can infect a variety of animal groups, including primates, carnivores, insectivores, and rodents. This ability to infect mammalian species that are not closely related may have implications for other wildlife species. It was first isolated from a captive pygmy marmoset that died from severe respiratory disease and was later isolated from a striped skunk in which it caused acute hepatitis and pneumonia. Infection in **PORCUPINES** affects the respiratory tract causing nasal and ocular discharge and pneumonia. Respiratory disease in infected captive African pygmy hedgehogs and a wild gray fox have also been seen.

**IN DEER**, Odocoileus adenovirus (OdAdV) causes adenovirus hemorrhagic disease (AHD) often resulting in rapid death, especially in fawns. See the disease fact sheet on AHD for details.

**OTARINE ADENOVIRUS** (OtAdV-1) causes diarrhea and acute hepatitis in California sea lions.

Adenovirus infections are an important source of economic loss to the poultry industry. **WILD BIRDS** generally do not exhibit illness from adenovirus infection, although respiratory disease or hepatitis may be seen in crowded captive populations. Examples of **AVIAN ADENOVIRUSES** causing disease in wild bird species held in captivity include respiratory signs in Northern Bobwhites, hepatitis in quail and pigeons, and enteritis and hepatitis in American kestrels. Wild ducks may be the source of the adenovirus that causes egg drop syndrome in domestic poultry, as supported by serologic evidence of antibodies in a variety of waterfowl species in many areas.

Adenoviruses are commonly found in captive **REPTILES**, especially bearded dragons. Infected animals may remain asymptomatic or exhibit anorexia and weight loss, weakness and abnormal postures, and sudden death. Adenoviruses have also been detected in apparently healthy wild bearded dragons in Australia.

Infected bearded dragons may remain healthy and shed the virus, potentially transmitting it to other bearded dragons. Testing of new animals for captive collections is important to prevent introduction of the virus and its spread.

**PREVENTION** Decreasing density among wild animals in captivity and in the wild near artificial feeding or water sources can help reduce transmission of adenoviruses. Proper disposal of deer carcasses and not moving infected, live deer to new areas is important to reduce spread of AHD.

Porcupines are an example of a species at risk for infection by skunk adenovirus due to the virus's unusual ability to infect a variety of unrelated mammalian species.

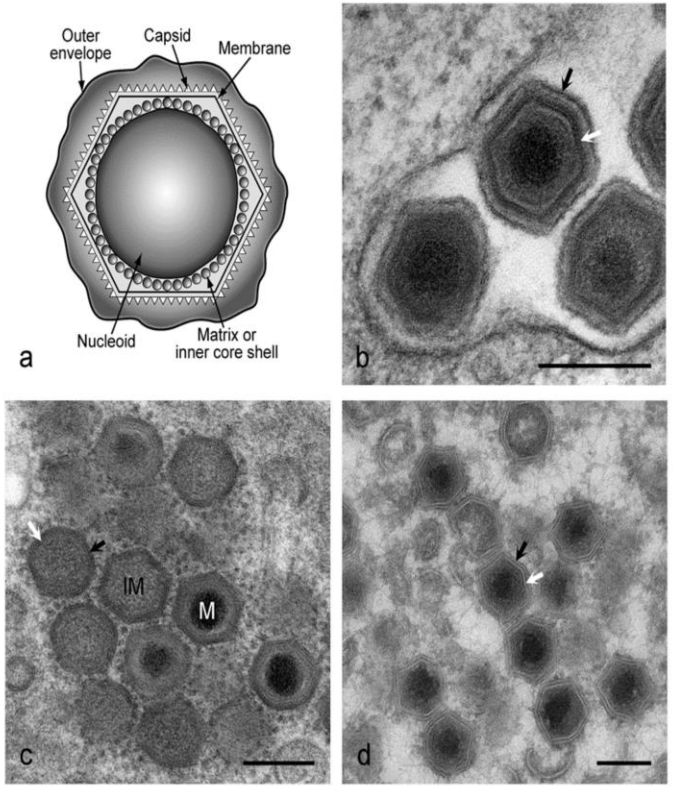
The NYS Wildlife Health Program | cwhl.vet.cornell.edu
A partnership between NYS Dept. of Environmental Conservation and Cornell Wildlife Health Lab
ALERT HOW WHO
MAR 2022

# Lab. diagnosis

- **Specimens:**
  - Depends upon nature of infections.
  - Throat swab, nasopharyngeal aspirates, conjunctival aspirates, conjunctival swab, urine, stool, blood, body fluids
- **Microscopy:** Electron microscope
- **Antigen detection**: ELISA, DFA test
- **Serology** or antibody detection
- **Molecular techniqu**e: PCR, DNA probe
- **Virus culture**: cell line culture

# Asfarviridae

- African swine fever virus is a large enveloped DNA virus that is the sole member of the genus Asfivirus within the family Asfarviridae (Asfar = African swine fever and related viruses).
- African swine fever virus is the only known DNA arbovirus and is transmitted by soft ticks of the genus Ornithodoros.
- Virus strains are distinguished by their virulence to swine, which ranges from highly lethal to subclinical disease.
- Strains can also be differentiated by their genetic sequences, and the virus-encoded p72 (also referred to as p73) gene can be used for genotyping the virus; however, the genomic diversity of the virus in nature remains to be thoroughly characterized.
- The genome of African swine fever virus contains a unique complement of multigene families.



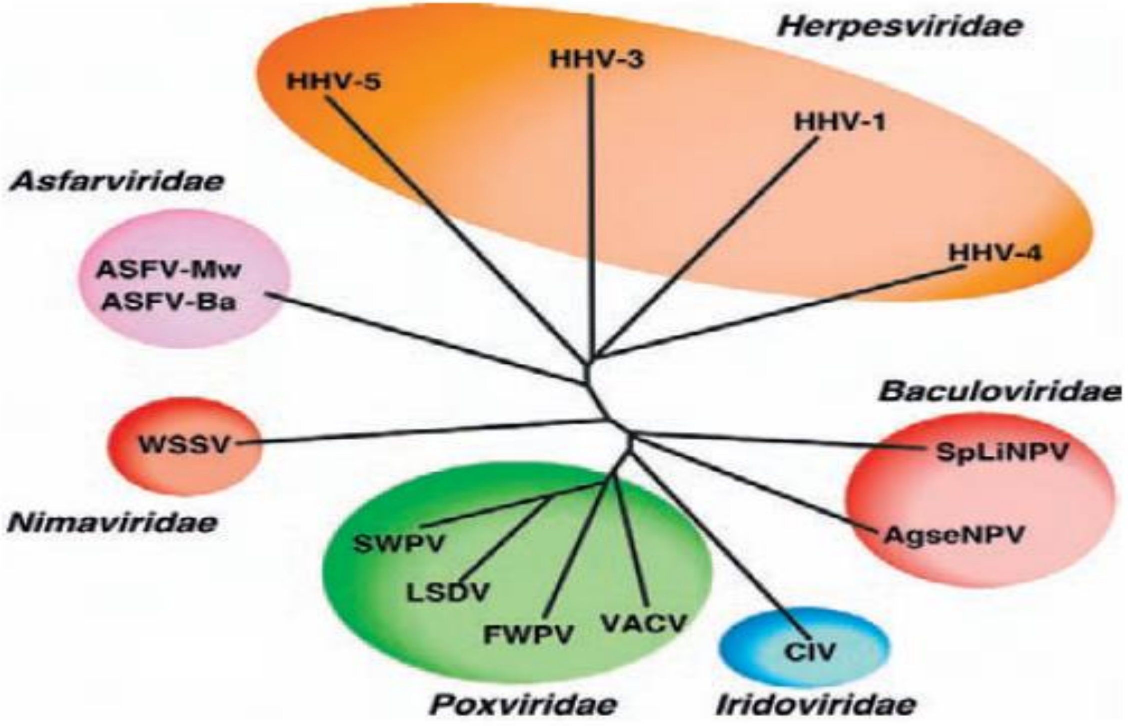



**FIGURE 8.1** Phylogenetic tree comparing the dUTPase proteins encoded by African swine fever virus (AFSV) with those of other DNA viruses. Sequences were aligned using ClustalW and the tree displayed using Treeview. Sequences shown are from: ASFV_Mw and -Ba, Malawi and Ba71V isolates of African swine fever virus, WSSV white spot syndrome virus, SWPV swinepox virus, LSDV lumpy skin disease virus, FWPV fowlpox virus, VACV vaccinia virus, CIV chilo iridescent virus, AgseNPV Agrotis segetum granulosis virus, SpLiNPV spodoptera liturna nucleopolyhedron virus, HHV-1 human herpesvirus 1, HHV-3 human herpesvirus 3, HHV-4 human herpesvirus 4, HHV-5 human herpesvirus 5. *(Provided by Dr. D. Chapman, IAH, Pirbright.) [From* Virus Taxonomy: Eighth Report of the International Committee on Taxonomy of Viruses *(C. M. Fauquet, M. A. Mayo, J. Maniloff, U. Desselberger, L. A. Ball, eds), p. 142. Copyright © Elsevier (2005), with permission.]*

# Viral Characteristics

- The virions - large, complex and in some structural respects resembles poxviruses.
- Enveloped, approximately 200 nm in diameter,
- Contain a complex icosahedral capsid, approximately 180 nm in diameter
- Virus consists of a nucleoprotein (70-100nm in diameter) surrounded by an icosahedral capsid and externally by a lipid layer.
- Genome linear, double stranded (170 - 190 kb in length)
- Encodes 150 - 200 proteins.
- Virus replicates in cytoplasm of host cells (swine macrophages *in vivo* and *in vitro*) and in soft ticks of the genus *Ornithodorus*.
- virus particles are stable in the environment being considerably resistant to heat and to a wide pH changes (several hours at pH 4 or pH 13).
- African swine fever virus is thermolabile and sensitive to lipid solvents.
- Survives for months and even years in refrigerated meat.

# **Virus survival** (*in the environment*)

- ASF virus, in a suitable protein environment, is stable over a wide range of temperature and pH.
- It survive in serum at room temperature for 18 months, in refrigerated blood for six years and in blood at 37°C for a month.
- Heating at 60°C for 30 minutes will inactivate the virus.
- In the absence of a protein medium, viability is greatly reduced.
- ASF virus is generally stable over a pH range of 4–10 but in a suitable medium (serum) has been shown to remain active at lower and higher values for between a few hours and three days.
- Putrefaction does not necessarily inactivate the virus, which may remain viable in faeces for at least 11 days, in decomposed serum for 15 weeks and in bone marrow for months.
- As a result of its tolerance to a wide range of environmental factors, only certain disinfectants are effective in the control of ASF.

# **Virus survival** (*in the host*)

- After infection with ASF virus, domestic pigs may shed infective amounts of virus for 24–48 hours before clinical signs appear.
- During the acute stage of disease, enormous amounts of virus are shed in all secretions and excretions and high levels of virus are present in tissues and blood.
- Pigs that survive the acute disease remain infected for several months but do not readily shed virus for more than 30 days.
- As in wild suids, infective levels of virus are found only in lymph nodes; other tissues are unlikely to contain infective levels of virus for more than two months after infection.
- The exact length of time over which infective levels of virus are maintained in lymphoid tissues in either wild suids or domestic pigs is unknown and is probably subject to considerable individual variation.

# Viral replication

- Primary isolates of African swine fever virus replicate in swine monocytes and macrophages.
- After adaptation, some isolates can replicate in certain mammalian cell lines.
- Replication occurs primarily in the cytoplasm, although the nucleus is needed for viral DNA synthesis and viral DNA is present in the nucleus soon after infection.

# Viral replication in host

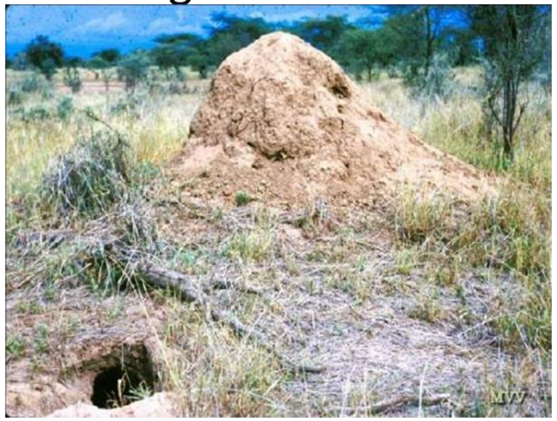
- Virus enters susceptible cells by receptor-mediated endocytosis
- After entry into the cytoplasm, virions are uncoated and their DNA is transcribed by a virion-associated, DNA-dependent RNA polymerase (transcriptase).
- DNA replication: parental genomic DNA serves as the template for the first round of DNA replication, the product of which then serves as a template for the synthesis of large replicative complexes that are cleaved to produce mature virion DNA.
- Late in infection, African swine fever virus produces arrays of virions in the cytoplasm.
- Infected cells form many microvillus-like projections through which virions bud out.

# The Disease - African Swine Fever (warthog disease)

- **DEFINITION:**
  - “ASF is a highly contagious viral disease of domestic pigs, ASF manifests itself as a haemorrhagic fever and results in up to 100 percent mortality.
  - The catastrophic effect of this disease on pig production, from household to commercial level, has serious socio-economic consequences and implications for food security.
  - It is a serious transboundary animal disease with the potential for rapid international spread”.

# Epidemiological features

• Warthog burrow



**Susceptible species**

- Only species of the pig family (*Suidae*) are susceptible to infection with ASF virus.
- Domestic pigs are highly susceptible to ASF, which shows no breed, age or sex preference.
- A high proportion of the pigs in these populations are serologically positive for ASF and apparently healthy

**Transmission**

- Transmitted by arthropods Soft ticks (Ornithodoros),which are likely vectors and act as reservoirs
- ASFV is transmitted from tick to tick sexually, transovarially and transstadially and sexually from males to females via the spermotheca.
- Infected ticks feeding on piglets transmit infective levels of virus in their saliva, which acts as an anticoagulant, to cause viraemia in the pigs sufficient to infect other ticks
- During an epizootic, transmission is by direct contact between infected pigs and their secretions and excretions.
- Infection generally occurs via the oronasal route.
- Aerosol transmission occur only over very short distances.
- Spread via fomites - contaminated vehicles, equipment and clothing - is likely when there are high levels of environmental contamination.
- Iatrogenic spread via contaminated needles

## Pathogenesis

- After infection by the oronasal route the virus replicates in the pharynx, tonsils and dependent lymph nodes.
- Viremia is followed by infection of bone marrow, lymph nodes, lungs, kidneys and liver where further replication takes place in cells of the lymphoreticular system.

**Clinical signs**

- The clinical symptoms of ASFV are very similar to classical swine fever virus and the two diseases normally have to be distinguished by laboratory diagnosis.
- Clinical disease is usually peracute or acute.
- Subacute or chronic manifestations of ASF may occur, particularly when less virulent strains are involved,

# Immunity

- Antibodies against ASF are detectable in serum 7–12 days after clinical signs appear and persist for long periods, possibly for life, in both warthogs and domestic pigs.
- They do not protect fully against subsequent infection in domestic pigs, although a degree of immunity to infection with homologous strains of virus has been reported.
- Serologically positive sows transmit antibodies to piglets in colostrum.
- In subacutely and chronically infected pigs, virus replication continues in the presence of antibodies.
- The deposition of immune complexes in tissue may account for many of the lesions observed in these forms of disease.
- There are no known serological cross-reactions with other viruses.

# Laboratory diagnosis

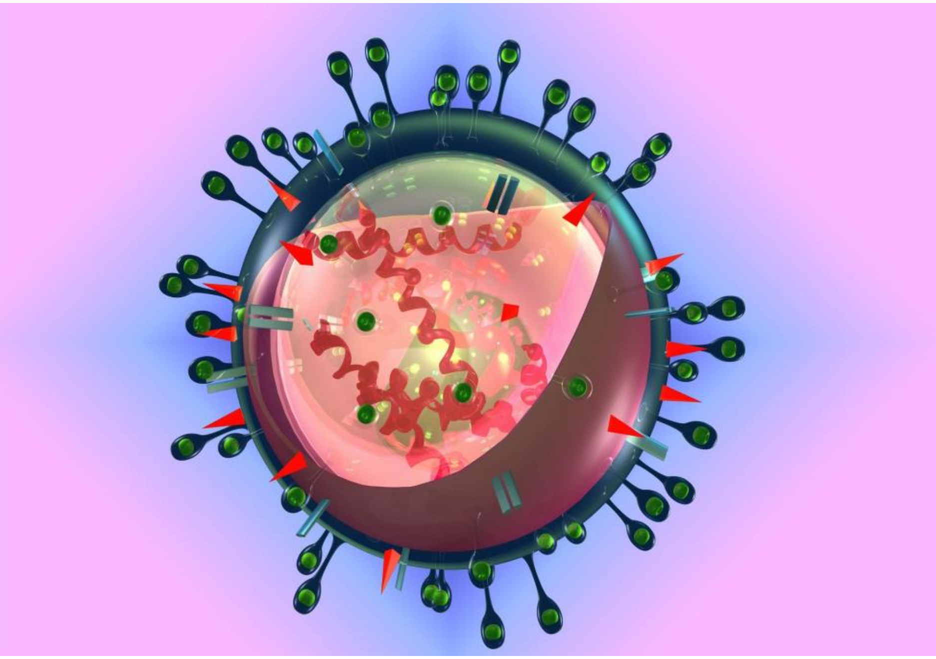
- Laboratory confirmation of a presumptive diagnosis of ASF depends upon detection of the virus or detection of antibodies.
- Since most pigs die of acute ASF before antibodies are produced, detection of the virus is the most important method of diagnosis.
- ***Collection and transport of diagnostic specimens.***
- Preferred samples for virus isolation/antigen detection are:
  - ✓ tissue samples from lymph nodes, spleen and tonsils collected aseptically and kept chilled but not frozen;
  - ✓ Whole (unclotted) blood collected aseptically into ethylenediamine tetra-acetic acid (EDTA) or heparin from febrile pigs up to five days after the onset of fever.

✓ **Virus isolation**

✓ **Antigen and antibody detection**

✓ **Virus's genetic material detection**

# RNA Viruses



**RNA Virus Family**

- 2.1. Rhabdoviridae
- 2.2. Picornaviridae
- 2.3. Retroviridae
- 2.4. Orthomyxoviridae
- 2.5. Paramyxoviridae
- 2.6. Bunyaviridae
- 2.7. Flaviviridae
- 2.8. Togaviridae
- 2.9. Reoviridae
- 2.10. Coronaviridae
- 2.11. Birnaviridae

# Picornaviridae

| Family | Genus | Virus |
|---|---|---|
| *Picornaviridae* | *Erbovirus* | Equine rhinitis B virus |
| | *Aphthovirus* | Bovine rhinitis A, B viruses<br>Equine rhinitis A virus<br>Foot-and-mouth disease virus, seven serotypes, many subtypes |
| | *Teschovirus* | Porcine teschovirus 1–11 (formerly porcine enteroviruses) |
| | *Enterovirus* | Swine vesicular disease virus<br>Porcine enterovirus B<br>Bovine enterovirus A, B (serotypes 1, 2) |
| | *Cardiovirus* | Encephalomyocarditis virus |
| | *Tremovirus* | Avian encephalomyelitis virus |
| | *Sapelovirus* | Porcine sapelovirus |
| | *Avihepatovirus* | Duck hepatitis A virus |
| | *Parechovirus* | Human parechovirus |
| | *Kobuvirus* | Aichi virus |
| | *Hepatovirus* | Hepatitis A virus |
| | *Senecavirus* | Seneca Valley virus |

**Figure 52.2** Classification of picornaviruses of importance.

# Properties of Picornaviruses

- An important difference between viruses in the various genera of picornaviruses is their stability at low pH.
- Specifically, the aphthoviruses are unstable below pH 7, whereas the enteroviruses, hepatoviruses, cardioviruses, and parechoviruses are stable at pH 3.
- All picornaviruses are single-stranded, positive-sense RNA viruses.
- Picornavirus virions are non-enveloped, ~ 30 nm in diameter, and have icosahedral symmetry.
- Virions appear smooth and round in outline.
- The genome consists of a single molecule of linear, positive-sense, ssRNA, 7–8.8 kb in size.
- Genomic RNA is infectious.
- The virions are constructed from 60 copies each of four capsid proteins, VP1 (also designated 1D), VP2 (1B), VP3 (1C) and VP4 (1A), and a single copy of the genome linked protein, VPg (3B) (*Mr variable).*
- VP1, VP2, and VP3 are structurally similar to one another, and differing primarily in the size and conformation of the loops.

# Cont…

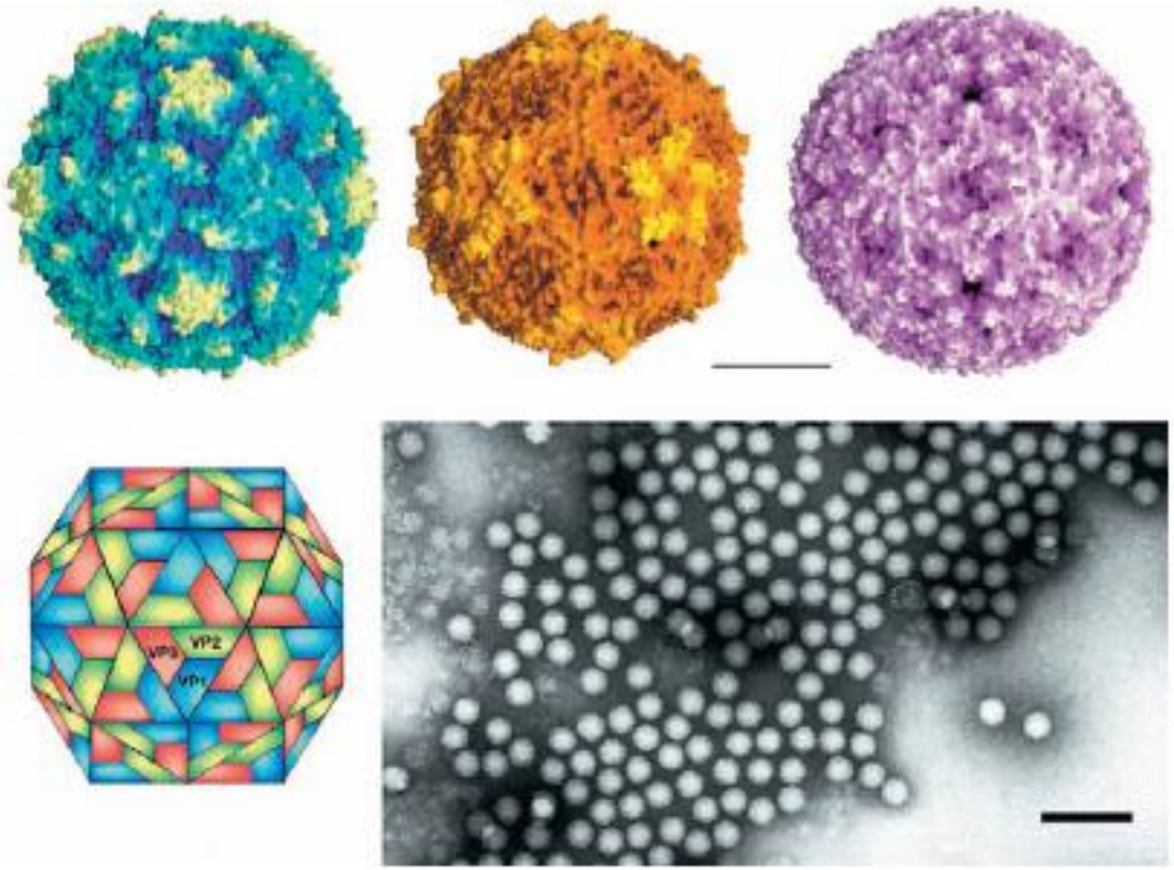
- The stability of picornaviruses to environmental conditions is important in the epidemiology of the diseases they cause, and in the selection of methods of disinfection.

- For example, if protected by mucus or feces and shielded from strong sunlight, most picornaviruses are relatively heat stable at usual ambient temperatures.

- Some enteroviruses, for instance, may survive for several days, and often weeks, in feces.

- Aerosols of aphthoviruses are less stable, but under conditions of high humidity they may remain viable for several hours.

- Because of differences in their pH stability, only certain disinfectants are suitable for use against each virus; ($NaCO_3$ for FMD but is not for SVD virus).




(Top) Pictures of picornavirus structures; poliovirus type 1 (PV-1) (Left), mengo virus (Center) and footand- mouth disease virus serotype O (FMDVO) (Right). The bar represents 10 nm. (Bottom left) Diagram of a picornavirus particle. The surface shows proteins VP1, VP2 and VP3. The fourth capsid protein, VP4, is located about the internal surface of the pentameric apex of the icosahedron. (Right) Negative contrast electron micrograph of poliovirus (PV) particles. The bar represents 100 nm.

# Virus Replication

- Poliovirus served as the basis for analyzing the replication pattern for all other.
- The cellular receptors for many picornaviruses are diverse.
- FMD virus can use two different receptors, but this change in receptor specificity results in attenuation of the virus.
- FMD viruses can also enter cells via Fc receptors if virions are complexed with non-neutralizing IgG molecules.
- The pathway(s) following attachment of the virus to its receptor and the release of the virion RNA into the cytoplasm of the host cell varies among the picornaviruses.
- For poliovirus, its interaction with the cell receptor induces structural changes in the virion.
- It is proposed that the RNA from the virus particle gains access to the cytoplasm of the host cells through this membrane pore.
- For FMD virus, the entry process is different, as the binding of this virus to its receptor does not induce changes in the virion structure, but simply functions as a docking mechanism; once bound to the receptor, the virus enters the cell through the endosomal pathway.

# Cont…

- Viral RNA synthesis takes place in a *replication complex*, which comprises RNA templates, the virus-coded RNA polymerase, and several other viral and cellular proteins, tightly associated with newly assembled smooth cytoplasmic membrane structures.
- The completed complementary strand in turn serves as a template for the synthesis of virion RNA.
- Most of the replicative intermediates found within the replication complex consist of a full-length complementary (negative-sense) RNA, from which several nascent plus sense strands are transcribed simultaneously by viral RNA polymerase.
- Picornaviruses do not have a defined mechanism of cellular exit, and large paracrystalline arrays accumulate in the infected cells.

# Diseases of animals caused by Picornavirus family

**Table 52.1** Picornaviruses of animals

| Virus | Genus | Host species | Significance of infection |
|---|---|---|---|
| Foot-and-mouth disease viruses | *Aphthovirus* | Cloven hoofed animals | Seven serotypes: O, Asia 1, A, C, SAT1, SAT 2, SAT 3. Many subtypes and numerous distinct antigenic strains recognized. Listed disease, highly contagious disease of international importance |
| Swine vesicular disease | *Enterovirus* | Pig | Mild vesicular disease, clinically indistinguishable from foot-and-mouth disease |
| Porcine sapelovirus (porcine enterovirus A, porcine enterovirus 8) | *Sapelovirus* | Pig | Infection is frequently asymptomatic. Associated with SMEDI |
| Porcine teschovirus 1 | *Teschovirus* | Pig | Varying virulence PTV-1 strains associated with encephalomyelitis. Highly virulent strains of PTV-1 (Teschen disease) confined to Eastern Europe and Madagascar. Worldwide distribution of mild PTV-1 strains (Talfan disease) |
| Porcine teschoviruses 2–11 | *Teschovirus* | Pig | Infection is frequently asymptomatic. Associated with encephalomyelitis, SMEDI, pneumonia and diarrhoea |
| Avian encephalomyelitis virus | *Tremovirus* | Chicken | Associated with avian encephalomyelitis in young chickens |
| Bovine enterovirus A, B | *Enterovirus* | Wide range of ruminant species | Widespread distribution. Isolated from normal cattle and from cattle with enteric, respiratory and reproductive disease problems |
| Equine rhinitis virus A | *Aphthovirus* | Horse | Common, systemic infection. Usually subclinical infection but also associated with respiratory disease |
| Encephalomyocarditis virus | *Cardiovirus* | Wide host range, rodents are natural hosts | Infection is often subclinical in pigs but associated with sudden death in young pigs and reproductive failure in sows |
| Bovine rhinitis B virus | *Aphthovirus* | Cattle | Widespread distribution. Isolated from normal cattle and from animals with respiratory disease, usually in association with other bovine respiratory viruses |
| Equine rhinitis B virus | *Erbovirus* | Horse | Widely distributed. Considered to be minor respiratory pathogen |

# Cont…

- FMD is caused by FMD virus that infects cloven-hoofed animals (e.g. cattle, pig, sheep, goats, water buffalo, camels).

- African buffalo are reservoir hosts and needs more research because of conflicting reports regarding the role of buffaloes.

- The FMD virus belongs to genus *Aphthovirus*, family *Picornaviridae*.

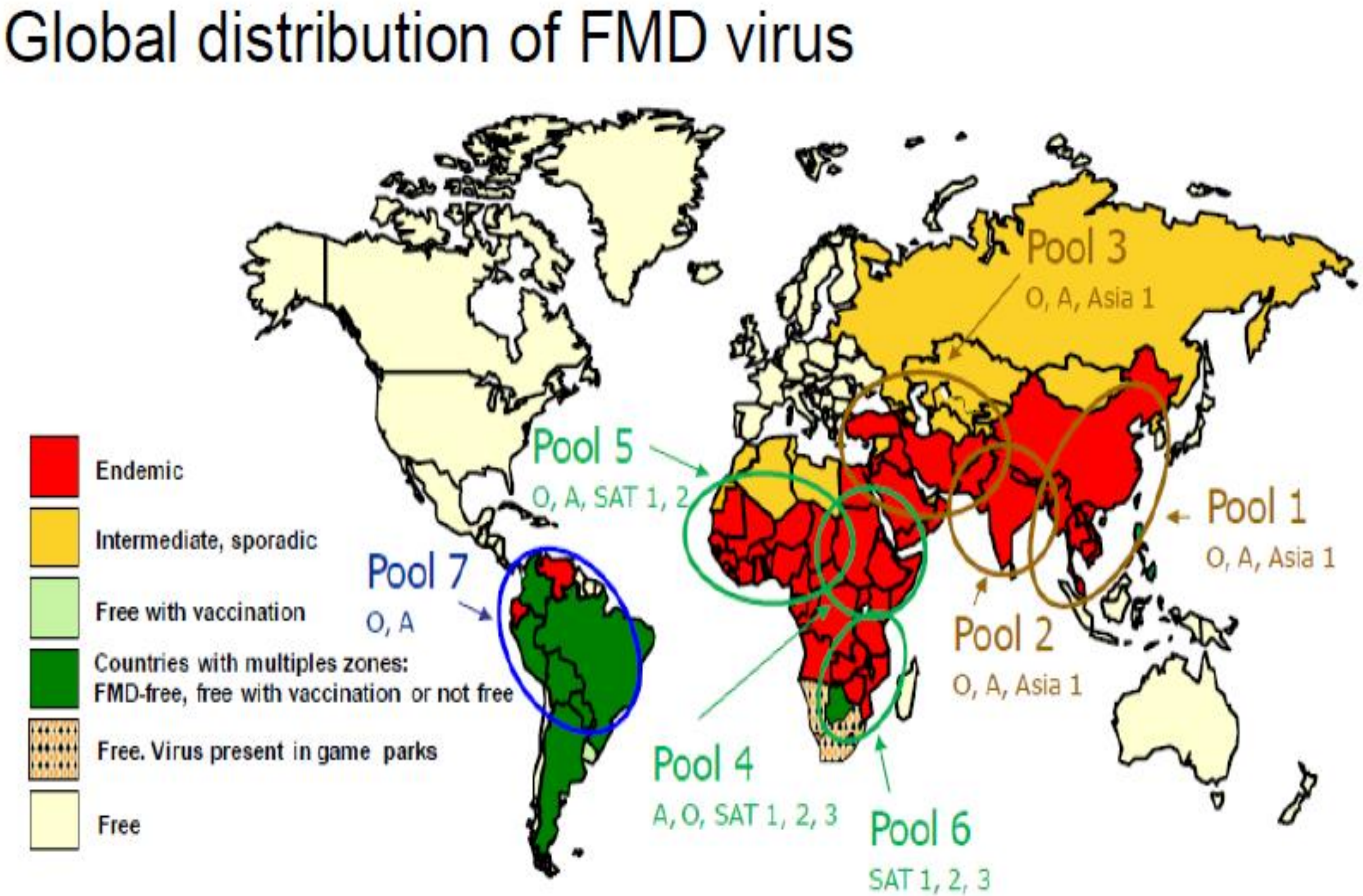
- Genus *Aphthovirus* has the following species
  - Rhinoviruses A and B (affects cattle).
  - Rhinovirus C (affects horses).
  - FMD virus (affects cloven-hoofed animals) and
  - FMD virus is type species of the *Aphthovirus* genus.

# Cont…

## The FMD virus

- FMD virus has seven serotypes (A, O, C, Asia 1, SAT1, SAT2 and SAT3).
- Some FMD virus serotypes has different topotypes e.g. serotype A has three topotypes: Asian, African and Euro-American.
- 
- Infection with one serotype does not provide protection against the other serotypes and even may not protect all of its subtypes. This has implication in FMD control as multiple vaccines are needed if more than one serotype is circulating in the country, e.g. 4 FMD serotypes in Ethiopia.
- FMD is highly contagious and economically important diseases. Because it has seven serotypes, laboratory diagnosis and serotype identification is critical for control measures to be instituted.

## Cont…




Pool positions are approximate and colours indicate that there are three principal pools, two of which can be subdivided into overlapping areas

**Figure Global distribution of FMD virus**

# Cont…

## FMD virus cont'd

- Infection with FMD virus leads to appearance of vesicles on the feet, in and around oral cavity, and mammary glands in animals.
- Mortality due to FMD is not common except in young animals that die from a multifocal myocarditis (seen macroscopically as "tiger heart").
- The FMD virus is single stranded, positive sense RNA virus.
- It is Icosahedral in shape, contains one open reading frame (ORF).
- The ORF encodes for a polyprotein: structural and non-structural proteins (12 proteins were known).
- The FMD virus is labile to low pH (<6.0).

# Cont…

## FMD virus genome organisation

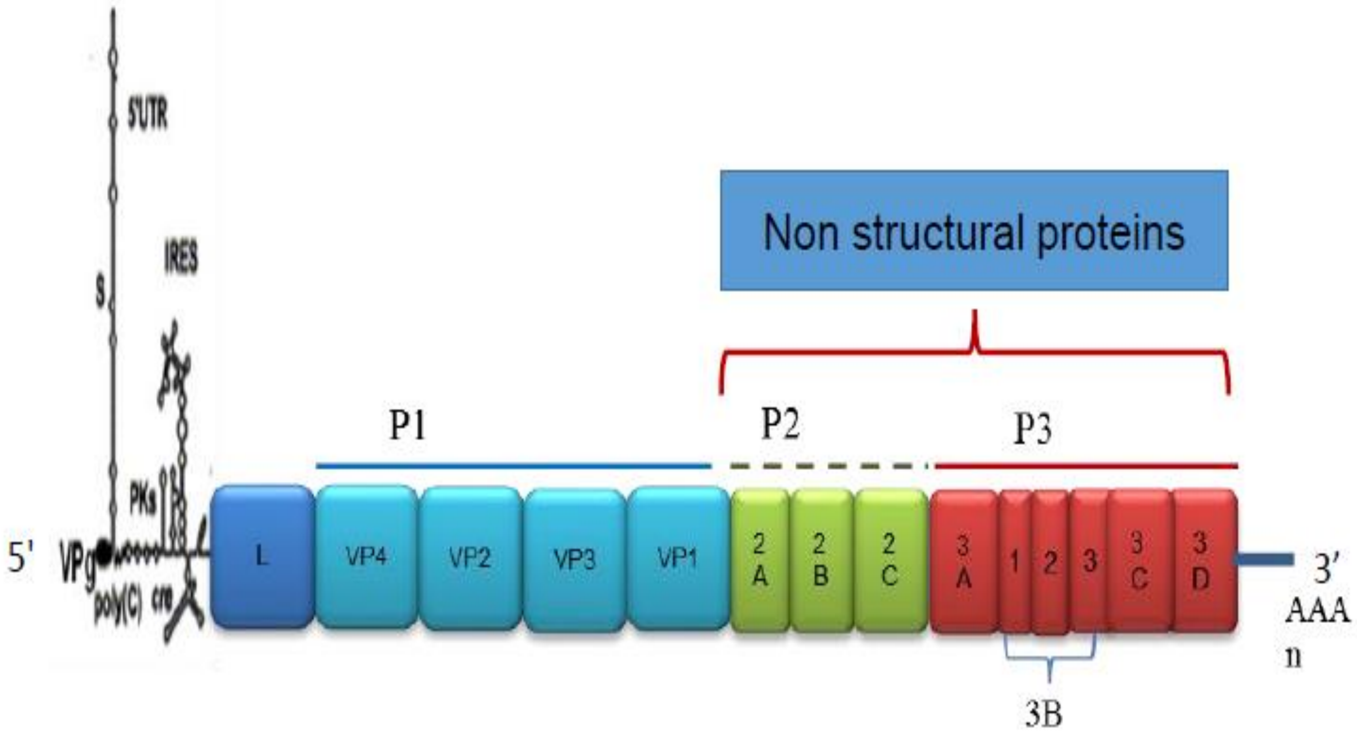



Figure. Unscaled genome organisation of the FMD virus

# Diagnosis of FMD

- **Samples**: epithelial tissue or vesicular fluid  or serum. At least 1 g of epithelial tissue should be collected from an unruptured or recently ruptured vesicle, usually from the tongue, buccal mucosa or feet.
- Epithelial samples should be placed in a transport medium (equal amounts of glycerol and 0.04 M phosphate buffer, pH 7.2-7.6, preferably with added antibiotics).
- Samples should be kept refrigerated or on ice until received by the laboratory.
- Sometimes in cattle oro-pharyngeal fluid can be taken by probing (sputum) cups in cattle or by swabbing the throat in pigs.
- Identification and isolation of the FMD virus need to be performed to exactly know the serotype circulating in the area or responsible for outbreak cases.

Cont…

## A. Identification of FMD virus

- 1. Virus isolation: Grind the tissue samples and make a 10% solution. Clarify by centrifuging on a bench at 2000x g for 10 minute.
- Innoculate onto cell cultures (BTY cells, IBRS2 cells, BHK21 cells) or into unweaned mice.
- The cell cultures should be examined for cytopathic effect (CPE) for 48 hours.
- 2. Immunological methods: Enzyme-linked immunosorbent assay (ELISA), Lateral flow device test,
- 3. Nucleic acid recognition methods: reverse transcription polymerase chain reaction (RT-PCR)-> run gel electrophoresis and then take forward PCR positive samples to sequencing. Finally perform molecular epidemiological analysis.
- Sometimes real-time RT-PCR can be used to quantify the virus in the sample.

# Cont…

## B. Serological tests used for FMD diagnosis

- B1. Virus neutralisation test:
- Known sera and unknown virus antigen with cell culture is needed.
- B2. Solid-phase competition ELISA (a prescribed test for international trade).
- B3. Liquid-phase blocking ELISA (a prescribed test for international trade).

- B4. Nonstructural protein (NSP) antibody tests:
- The tests commonly used are indirect ELISA and immunoblotting.
- Different purified non-structural proteins (3A, 3B, 3C, 3AB, 3ABC etc) can be used for this purpose.
- Some of the indirect ELISA test using purified non-structural proteins can be used as DIVA (differentiation of infection from vaccination) test.

# Control of FMD

- Different control measures can be used including vaccination, biosecurity measures, and test-and slaughter policy.
- The control measures differ between developed and developing countries.
- In developed countries, test-and-slaughter policy and biosecurity measures are used.
- In developing countries, vaccination using matching vaccine for serotypes circulating in the country is used.
- Vaccine matching is assessing the protection potential of existing or new vaccine strains against circulating field viruses.
- The vaccine can be monovalent, bivalent and polyvalent.
- Two vaccine producing plants are available in East Africa: National Veterinary Institute (Ethiopia) and KEVEVAPI (Kenya).

# Rhabdoviridae

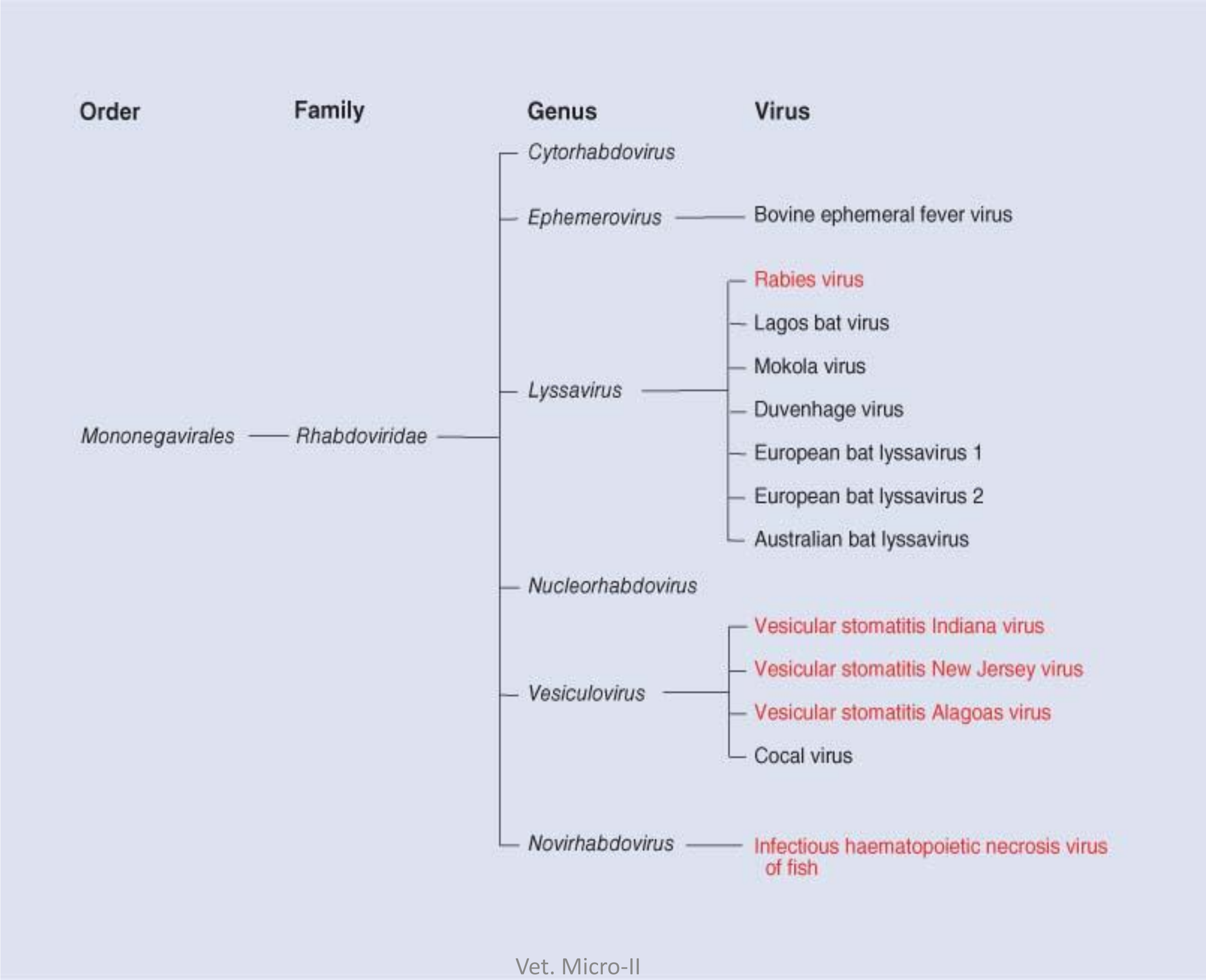
- Members of the family *Rhabdoviridae* (Greek *rhabdos*, rod) have characteristic rod shapes.
- The family *Rhabdoviridae* encompasses six genera of viruses that infect a broad range of hosts, including mammals, birds, fish, insects, and plants with some of the family members being transmitted by arthropod vectors.
- Rabies is one of the oldest and most feared diseases of humans and animals, Perhaps the most lethal of all infectious diseases.
- Vesicular stomatitis of horses, cattle, and swine was a significant problem in artillery and cavalry horses during the American Civil War.
- Bovine ephemeral fever was first recognized in Africa, and is now known to be widespread across Africa, most of southeast Asia, Japan, and Australia.
- More recently, a number of rhabdoviruses in at least two different genera have been identified as the cause of serious losses in the aquaculture industries of North America, Europe, and Asia.

# Properties of Rhabdoviruses

## Classification

- The family *Rhabdoviridae* includes four genera that contain animal viruses: the genera *Lyssavirus, Vesiculovirus, Ephemerovirus, and Novirhabdovirus.*
- Individual species of rhabdovirus are distinguished genetically and serologically.
- The genus *Lyssavirus* includes rabies virus and closely related viruses, including Mokola, Lagos bat, Duvenhage, European bat lyssaviruses 1 and 2, and Australian bat lyssavirus.
- Each of these viruses is capable of causing rabies-like disease in animals and humans.
- Certain terrestrial mammals are reservoir hosts of rabies virus, and bats are potential reservoirs of both rabies and the rabies-like viruses.

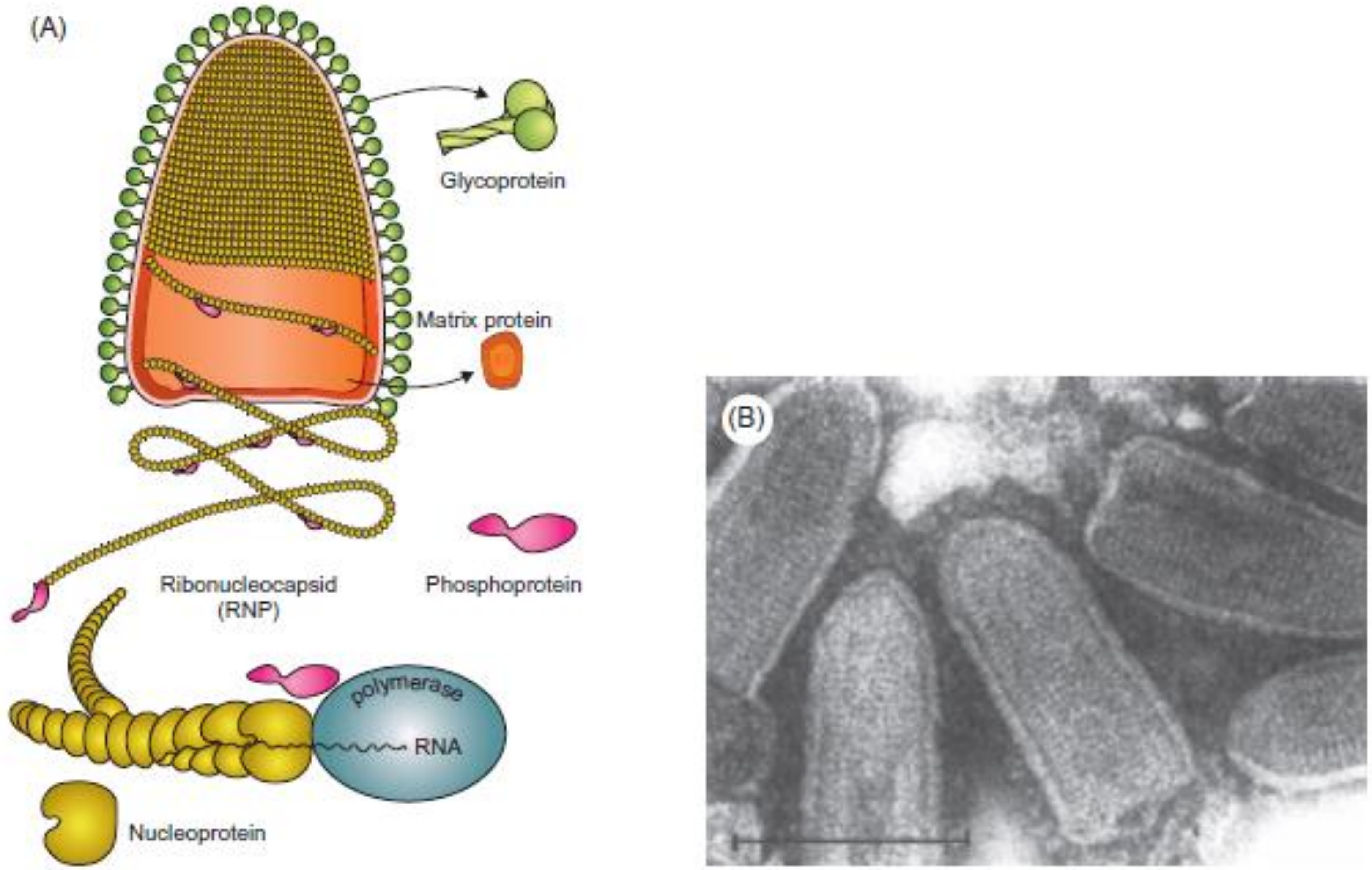
# Cont…



Order
Family
Genus
Virus
Mononegavirales
Rhabdoviridae
Cytorhabdovirus
Ephemerovirus
Bovine ephemeral fever virus
Lyssavirus
Rabies virus
Lagos bat virus
Mokola virus
Duvenhage virus
European bat lyssavirus 1
European bat lyssavirus 2
Australian bat lyssavirus
Nucleorhabdovirus
Vesiculovirus
Vesicular stomatitis Indiana virus
Vesicular stomatitis New Jersey virus
Vesicular stomatitis Alagoas virus
Cocal virus
Novirhabdovirus
Infectious haematopoietic necrosis virus of fish

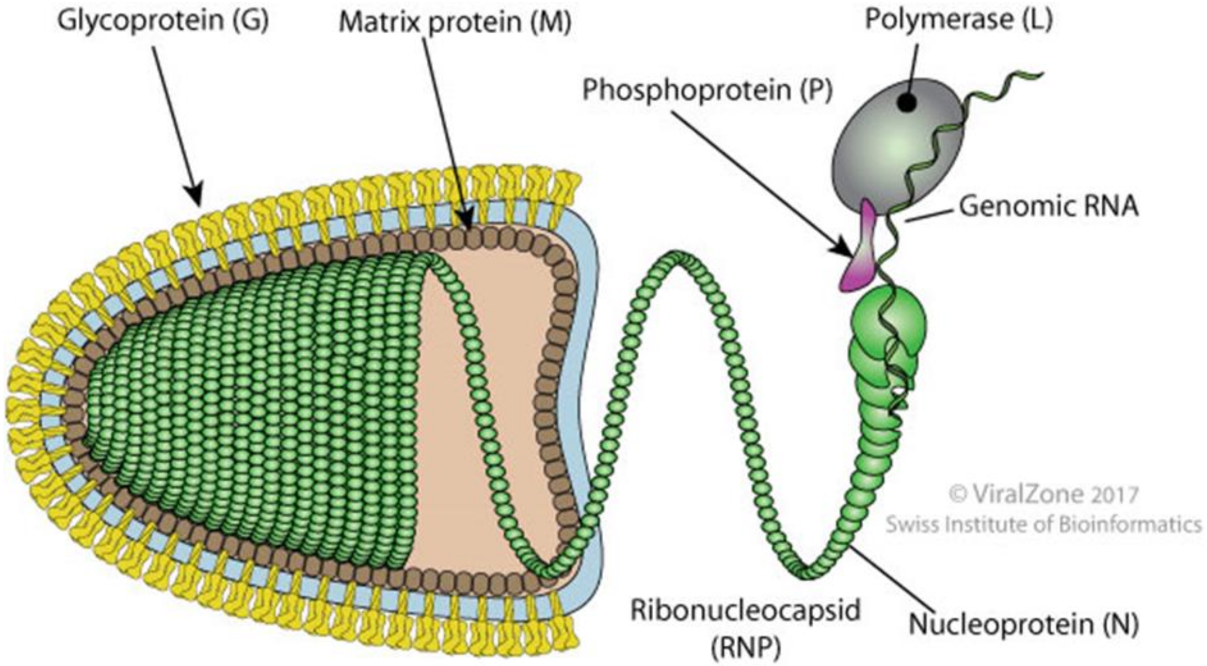
# Virion Properties

- Rhabdovirus virions are approximately 45–100 nm in diameter and 100–430 nm long, and consist of a helically coiled cylindrical nucleocapsid surrounded by an envelope with large (5–10 nm in length) glycoprotein spikes.
- The precise cylindrical form of the nucleocapsid is what gives the viruses their distinctive bullet or conical shape.
- The genome is a single molecule of linear, negativesense, ssRNA, 11–15 kb in size.
  - N is the nucleoprotein gene that encodes the major component of the viral nucleocapsid;
  - P is a cofactor of the viral polymerase;
  - M is an inner virion protein that facilitates virion budding;
  - G is the glycoprotein that forms trimers that make up the virion surface spikes;
  - L is the RNA-dependent RNA polymerase that functions in transcription and RNA replication.
- Some rhabdoviruses have additional genes or pseudogenes interposed.
- Rhabdoviruses are relatively stable in the environment, especially when the pH is alkaline
- Vesicular stomatitis viruses can contaminate water troughs for many days—but the viruses are thermolabile and sensitive to the ultraviolet irradiation of sunlight.
- Rabies and vesicular stomatitis viruses are inactivated readily by detergent-based disinfectants,
- Iodine-containing preparations are commonly applied as disinfectants for reducing or eliminating fish rhabdoviruses such as those that occur on the surface of living fish eggs.

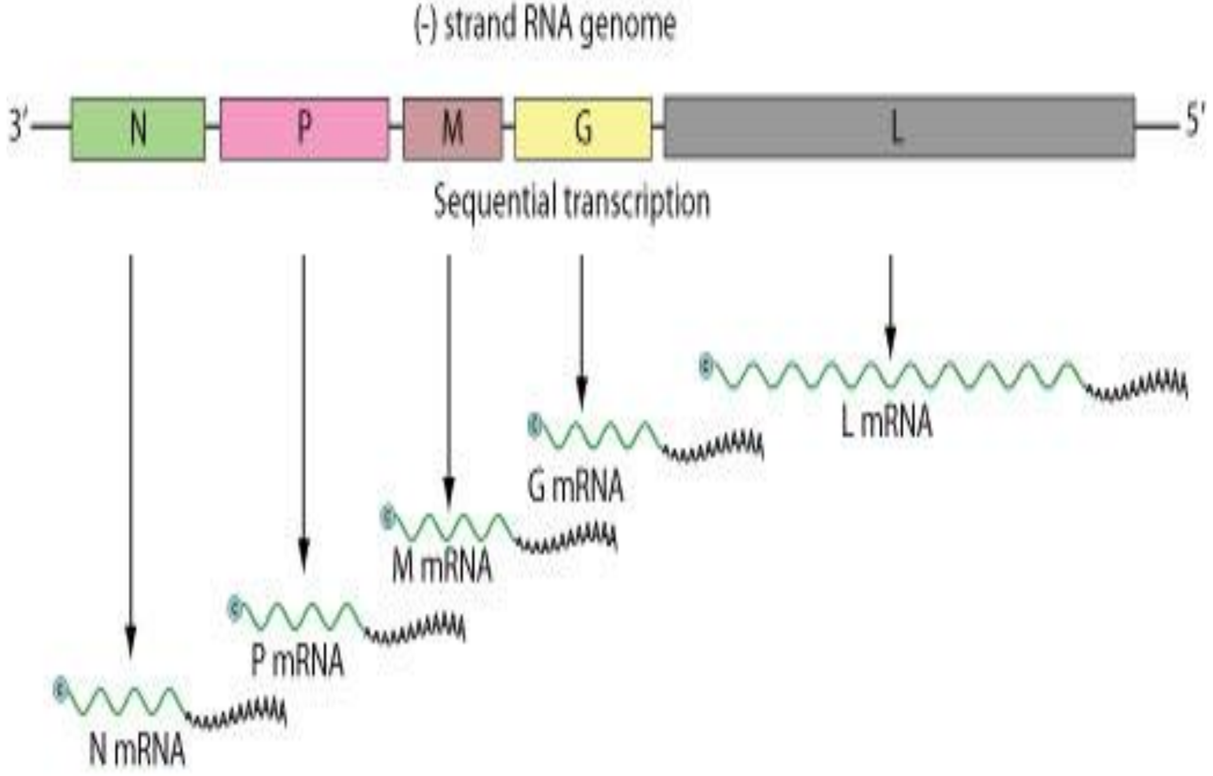



Family *Rhabdoviridae. (A) Diagram illustrating a rhabdovirus virion and the nucleocapsid structure (courtesy of P. Le Merder).* (B) Vesicular stomatitis Indiana virus showing characteristic bullet-shaped virions.

Virion: Enveloped, bullet shaped. 180 nm long and 75 nm wide. Certain plant rhabdoviruses are bacilliform in shape and almost twice the length.

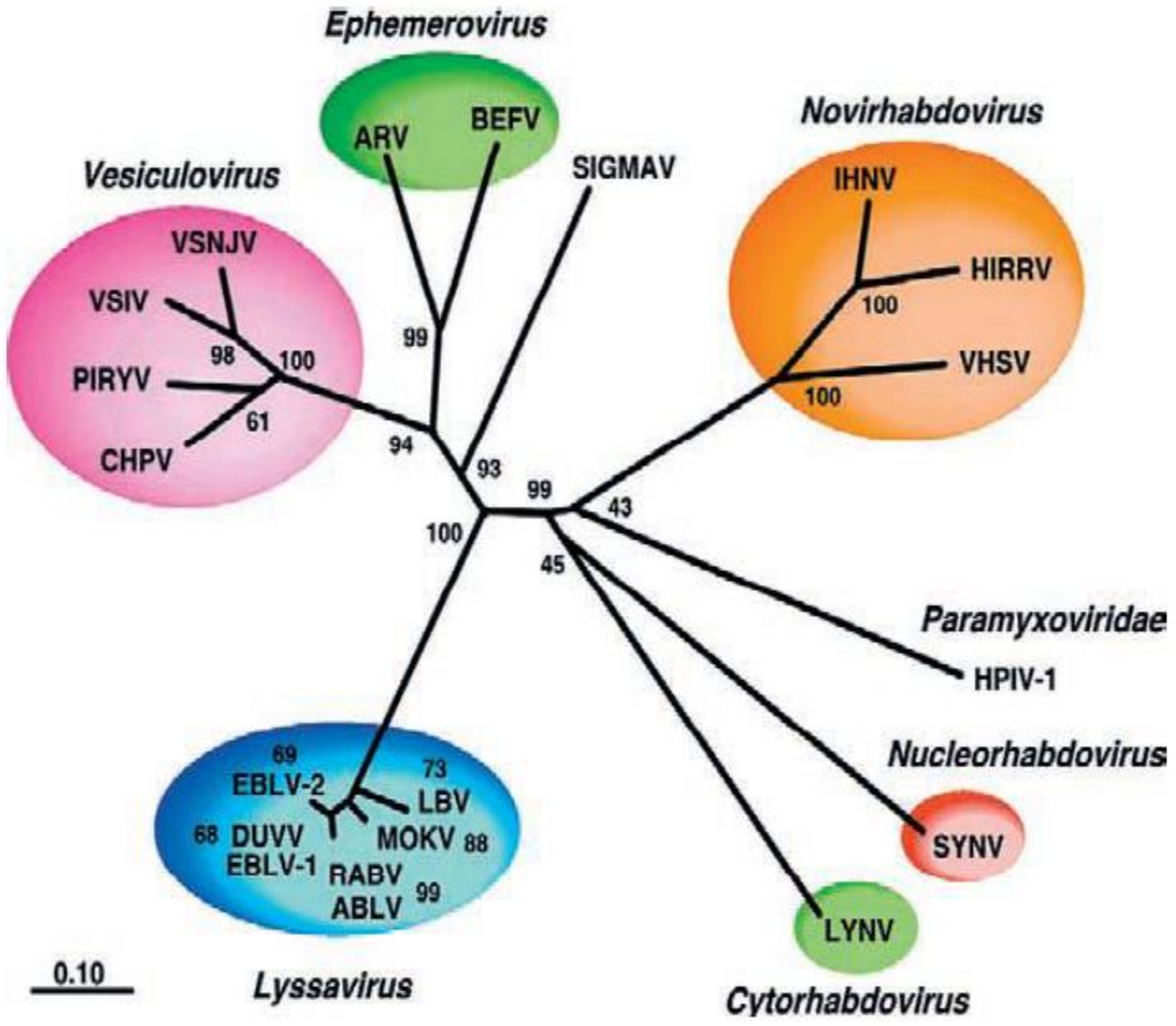


Negative-stranded RNA linear **genome**, about 11-15 kb in size. Varicosavirus genomes consists in two segments. Encodes for five to six proteins.

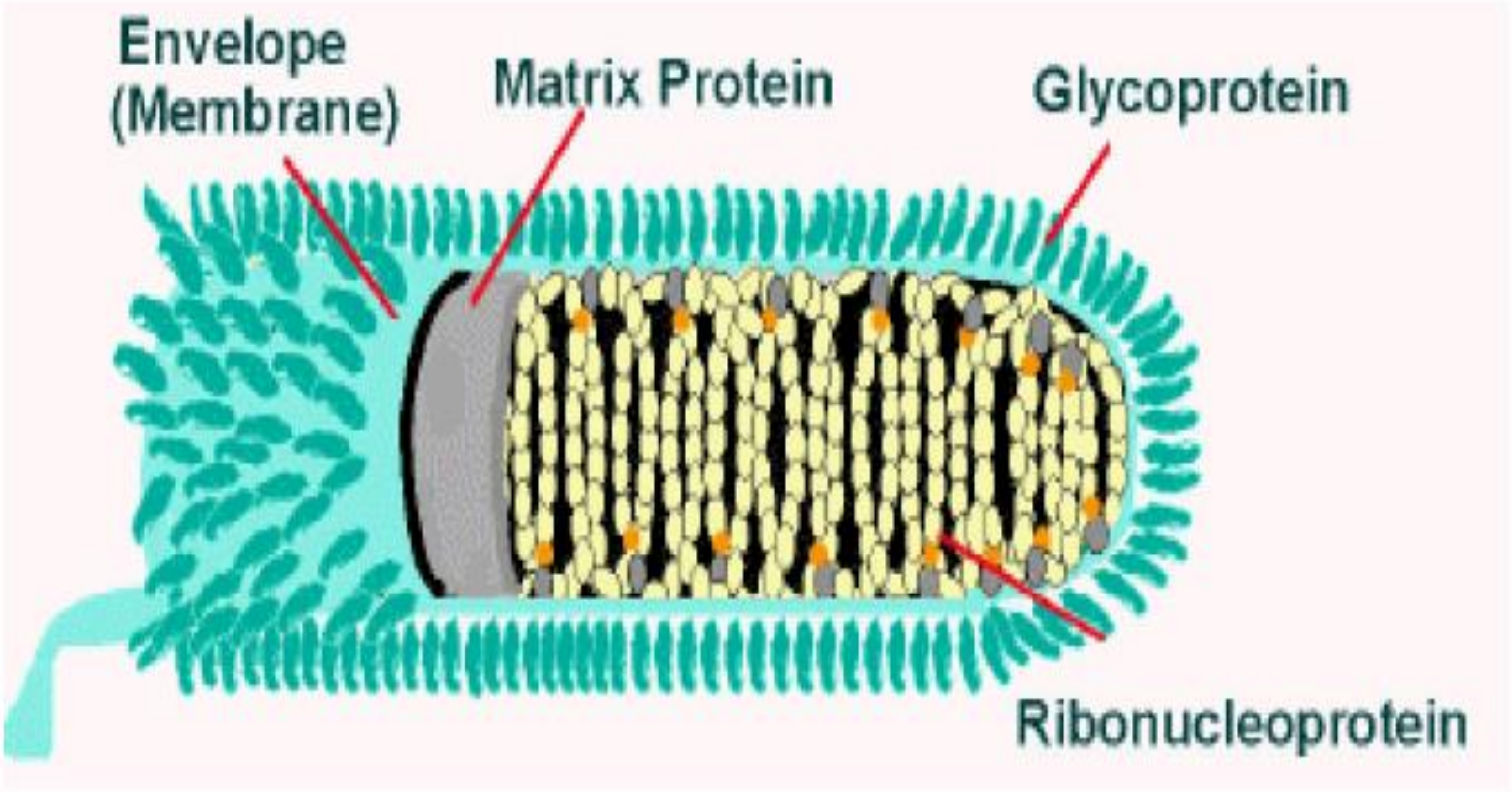


# Virus Replication

- Virus entry into host cells occurs by receptor-mediated endocytosis via coated pits, and subsequent pH-dependent fusion of the viral envelope with the endosomal membrane releases the viral nucleocapsid into the cytoplasm, where replication exclusively occurs.
- The viral glycoprotein G is solely responsible for receptor recognition and cell entry.
- Specific cell receptors have not clearly been identified for individual rhabdoviruses.
- Replication first involves mRNA transcription from the genomic RNA via the virion polymerase.
- When sufficient quantities of the nucleocapsid (N) and phosphoprotein (P) have been expressed, there is a switch from transcription of mRNA to positive-sense antigenomes, which then serve as the template for synthesis of negative-stranded, genomic RNA.




Family *Rhabdoviridae*: phylogenetic relationships based on a relatively conserved region of the N protein. Tree branch lengths are proportional to genetic distances




Structure of rabies virus (Source: CDC)

# Clinical infections

- Rhabdoviruses of veterinary importance are presented in following tables.
- They can be transmitted by bites of mammals, arthropod vectors or direct contact.
- Infection may also be acquired through environmental contamination.
- The best known and most important member of the *Rhabdoviridae* is rabies virus, a *Lyssavirus* (Greek *lyssa*, rage or fury). A number of distinct *Lyssavirus* genotypes produce clinical signs indistinguishable from rabies .
- More than 25 viruses isolated from animals have been classified in the genus *Vesiculovirus*.
    - The most important vesiculoviruses which infect domestic animals are the vesicular stomatitis Indiana virus and the vesicular stomatitis New Jersey virus.
- Bovine ephemeral fever virus, of significance in some countries, is the type species of the genus *Ephemerovirus*.
- Some fish diseases, such as infectious haematopoietic necrosis, viral haemorrhagic septicaemia and spring viraemia of carp are also caused by rhabdoviruses.

# Cont…



glycoprotein as antigen, are useful screening tests to determine if vaccinated cats and dogs have seroconverted.

Table 73.1 Lyssaviruses which cause classical rabies and rabies-like disease.

| Virus | Phylogroup | Genotype | Serotype | Geographical distribution | Comments |
|---|---|---|---|---|---|
| Rabies virus | 1 | 1 | 1 | Apart from Australia and Antartica, rabies virus (genotype 1) occurs on all continents. Many island countries are free of the disease | Causes fatal encephalitis in many mammalian species. Transmitted by wildlife species including foxes, racoons and bats; domestic carnivores also involved in transmission. Rabies is a major zoonotic disease with more than 50,000 human fatalities worldwide each year |
| Lagos bat virus | 2 | 2 | 2 | Africa | Isolated initially from fruit bats; also isolated from domestic animals with encephalitis |
| Mokola virus | 2 | 3 | 3 | Africa | Isolated initially from shrews; also isolated from domestic animals. Human infection reported |
| Duvenhage virus | 1 | 4 | 4 | Africa | Originally isolated from a human bitten by an insectivorous bat; additional cases reported in humans. Not reported in domestic animals |
| European bat lyssavirus 1 | 1 | 5 | – | Europe | Identified with increasing frequency in insectivorous bats. Human infection reported |
| European bat lyssavirus 2 | 1 | 6 | – | Europe | Isolated initially from a human with symptoms of rabies; present in insectivorous bats. Additional human cases reported; not reported in domestic animals |
| Australian bat lyssavirus | 1 | 7 | – | Australia | Identified in fruit bats and in insectivorous bats; human infection reported |

Table 73.2 Viruses of veterinary significance in the genera *Vesiculovirus* and *Ephemerovirus*.

| Genus, virus | Hosts | Comments |
|---|---|---|
| *Vesiculovirus* | | |
| Vesicular stomatitis Indiana virus | Cattle, horses, pigs, humans | Causes febrile disease with vesicular lesions; resembles foot-and-mouth disease clinically. Occurs in North and South America |
| Vesicular stomatitis New Jersey virus | Cattle, horses, pigs, humans | Causes febrile disease with vesicular lesions; infection more severe than that caused by the Indiana virus. Occurs in North and South America |
| Vesicular stomatitis Alagoas virus (Brazil virus) | Horses, mules, cattle, humans | Originally isolated from mules in Brazil |
| Cocal virus (Argentina virus) | Horses | Isolated initially from mites in Trinidad; occurs in South America |
| *Ephemerovirus* | | |
| Bovine ephemeral fever virus | Cattle | Causes febrile illness of short duration; occurs in Africa, Asia and Australia |

# Rabies

**Physical, Chemical, and Antigenic Properties (of the Etiologic Agent)**

- The rabies virion, approximately 180nmin length by 80nmin width, contains a single-stranded negative-sense genome RNA that is encapsidated with the nucleocapsid (N) protein, the RNA polymerase (L), and polymerase cofactor phosphoprotein (P).
  - This ribonucleoprotein (RNP) core, along with the matrix protein (M), is condensed into the typical bullet-shaped particle characteristic of rhabdoviruses.
- Glycoprotein (G) spikes are anchored into the RNP-M structure and protrude through the virus envelope.
- Antibodies produced against the G protein after vaccination with cell culture rabies vaccines induce the production of protective antibodies and function to neutralize the virus.
- The N protein is the most highly conserved of all of the viral proteins but actually has a high degree of diversity within short segments.
- As a result, molecular analyses of the N protein have provided information on identifying rabies virus genotypes that can be used as a molecular epidemiological surveillance tool to explain the evolution of rabies viruses and to track movement of rabies over time.
- For instance, when rabies is detected in animals in a previously rabies-free zone, the virus can often be traced back to the point of origination.

## Cont…

**Resistance to Physical and Chemical Agents.**

- Rabies virus is readily destroyed through exposure to soap and other disinfectants, desiccants, and sunlight.
- The virus is inactivated by heating to 56 ◦C for 30 min or by various diluted chemicals and disinfectants including a 0.1% bleach solution and a1%formalin solution.
- Saliva from a rabid animal is no longer considered infective in the environment or on intact skin after it has dried.
- In the case of human exposure, all wounds should be washed vigorously with soap and water for at least 15 min in order to inactivate the virus.

# Cont…

## Disease

- Rabies is a progressive encephalitic disease that almost always results in the death of the victim although rare cases of prolonged life and the survival of one human without the administration of vaccine has been reported using the Milwaukee protocol.
- In humans, early symptoms of rabies are indeterminate, flu-like and may include general malaise, headache, fever, and weakness or discomfort.
- These signs progress into more specific symptoms and may include insomnia, anxiety, confusion, partial paralysis, hypersalivation, hydrophobia, hallucination, and painful spasms of the pharyngeal muscles.
- Pain or itching at the site of the initial exposure wound (paresthesia) may also occur.
- Without intensive medical support, death usually occurs within a few days of the onset of these symptoms.

# Cont…

- Two clinical forms of the disease are recognized: furious rabies, and dumb (or paralytic) rabies.
- In the furious form:
- - the animal becomes restless, nervous, aggressive, and often extremely dangerous as it loses fear of humans and bites at anything that gains its attention.
- -the animal often cannot swallow water because of pharyngeal paralysis, giving rise to the old name for the disease, “hydrophobia”.
- -other characteristic signs include excessive salivation, exaggerated responses to light and sound, and hyperesthesia (animals commonly bite and scratch themselves).
- As encephalitis progresses, fury form lead to paralysis. Similar signs follow with convulsive seizures, coma, and respiratory arrest, with death occurring 2–14 days after the onset of clinical signs.

- Diagnosis:
- Samples: brain samples specifically from thalamus, cerebral cortex and medulla oblongata. Generally samples should be collected from at least two different parts of the brain in the laboratory.
- In the field, such sampling is not practical. Occipital foramen & retro-orbital route for brain sampling are used.
- The diagnosis is then conducted by
  - direct florescent antibody,
  - immunohistochemical staining to demonstrate rabies virus antigen in frozen sections of brain tissue (medulla, cerebellum, and hippocampus),
  - mouse inoculation technique,
  - tissue culture, and
  - Reverse transcription (RT) - polymerase chain reaction.
- Negri bodies in the neurons of affected animals are characteristic of rabies, but are difficult to identify.

# Cont…

- The most common request is to determine whether an animal known to have bitten a human is rabid or not.

- If rabies is suspected, the suspect animal must be killed and brain tissue collected for testing.

- Post-mortem diagnosis most commonly involves:
  - direct immunofluorescence or
  - immunohistochemical staining to demonstrate rabies virus antigen in frozen sections of brain tissue (medulla, cerebellum, and hippocampus).

- Molecular techniques - RT-PCR based on N gene.

# Cont…

- **Serological tests**:
  - VN in cell culture;
  - Fluorescent antibody test; and
  - Indirect ELISA.
- **Control is by Vaccination**

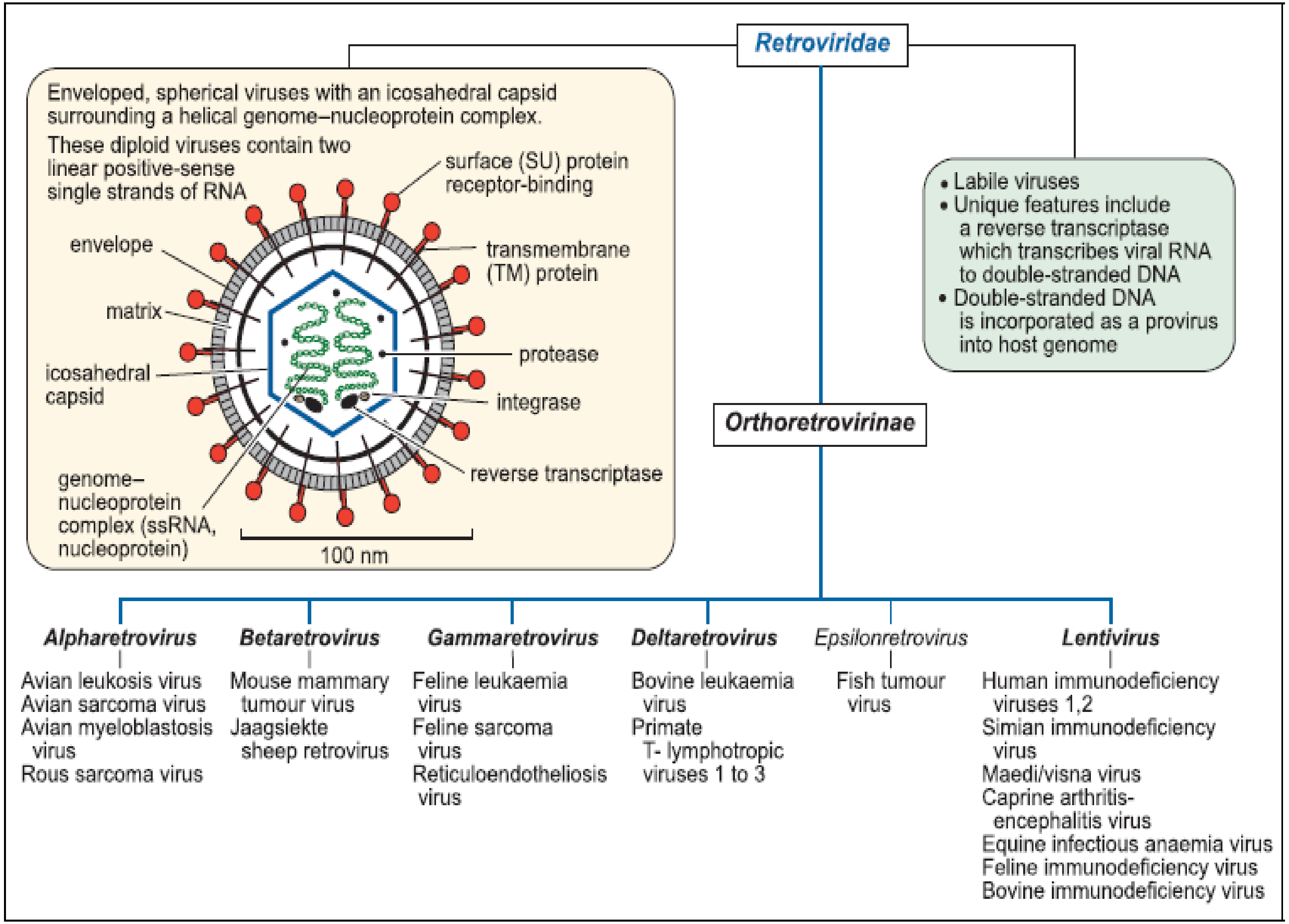
- Post exposure vaccination is given in humans but not in animals.
- Three kinds of vaccines are available: inactivated (killed), live-attenuated or biotechnology-derived.
- Parenteral vaccines are recommended for domestic animals and oral vaccines are used in wild animals.
- Rabies vaccines produced in compliance with OIE requirements protect against all variants of rabies virus including other phylogroup 1 lyssaviruses.

- **Control of Rabies**:
- In humans: thorough cleansing of the wound by immediate vigorous washing and flushing with soap and water is crucial.
- If the individual has been scratched or bitten or skin abrasions present, treatment is recommended.
- *Pre-exposure vaccination* of veterinarians and other individuals occupationally or otherwise at risk is important.
- The *Pre-exposure vaccination* is recommended for any one who will be at continual, frequent or increased risk of exposure.
- *Post-exposure prophylaxis consists of hyperimmune globulin* and a course of vaccine. The indication for post-exposure prophylaxis depends on the **type of contact** with the suspected rabid animal:

# Cont…

- Category I – touching or feeding animals, licks on intact skin - no exposure.
- Category II–nibbling of uncovered skin, minor scratches or abrasions without bleeding.
- Category III–single or multiple transdermal bites or scratches, contamination of mucous membrane with saliva from licks, licks on broken skin, exposures to bats.
- **In animals such as dogs:**
- Modified live virus vaccines: SAG2, SAD B19.
- Live recombinant vaccines: HAVR, -VRG,
- Inactivated vaccines, and recombinant (plant) viruses: Only attenuated or recombinant vaccines have been shown to be efficacious by the oral route in dogs but inactivated immunogens or recombinant antigens have shown to be efficacious orally in other carnivores.

# Retroviridae



## Cont…

- Retroviruses (Latin *retro*, backwards) are labile enveloped RNA viruses, 80 to 100 nm in diameter.
- The family name refers to the presence in the virion of a reverse transcriptase which is encoded in the viral genome.
- Reverse transcriptase acts as an RNA dependent DNA polymerase, which transcribes RNA to DNA.
- Under the influence of the reverse transcriptase, double-stranded DNA copies of the viral genome are synthesized in the cytoplasm of the host cell.
- During this process, repeat base sequences called long terminal repeats (LTR), containing several hundred base pairs, are added to the ends of the DNA transcripts.
- Transcripts are integrated into the chromosomal DNA as provirus, through the action of viral integrase.
- Integration occurs at random sites in the DNA and the sites of proviral integration determine the extent and nature of cellular changes.
- The LTR contain important promoter and enhancer sequences. Infectious viruses have four main genes: 5′-*gag*-*pro*-*pol*-*env*-3′ as illustrated.
- Because errors are relatively frequent during reverse transcription, a high mutation rate is a feature of retroviral replication.
- Recombination between retroviral genomes in doubly infected cells can occur due to transfer of reverse transcriptase from one RNA template to another.
- Consequently, quasi species frequently form and definitive classification at the level of species or below often proves difficult.

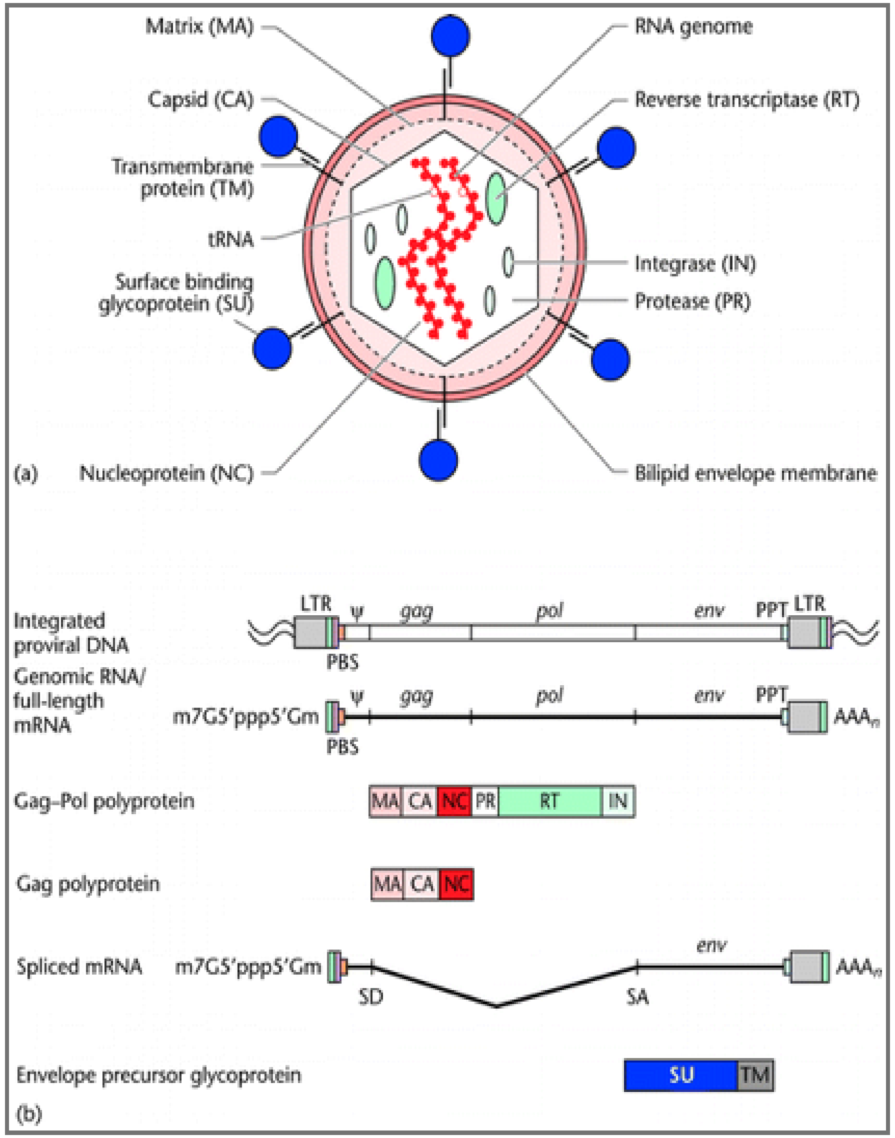



**Figure 1** Retroviral particle and genome structure. (a) Retrovirus particle showing the approximate location of its components using the standardized two-letter nomenclature for retroviral proteins. (b) Genome organization and gene expression pattern of a simple retrovirus, showing the structure of an integrated provirus linked to flanking host cellular DNA at the termini of its LTR sequences (U3-R-U5) and the full-length RNA that serves as genomic RNA and as mRNA for translation of the *gag* and *pol* ORFs into polyproteins. *env* mRNA is generated by splicing and encodes an Env precursor glycoprotein. LTR, long terminal repeat (U3-R-U5 for proviral DNA, derived from R-U5 downstream of 5′ cap and U3-R upstream of 3′ poly(A) in genome RNA); PBS, primer binding site; Ψ, packaging signal; PPT, polypurine tract; SD, splice donor site; SA, splice acceptor site.

# Reverse transcriptase of genomic RNA

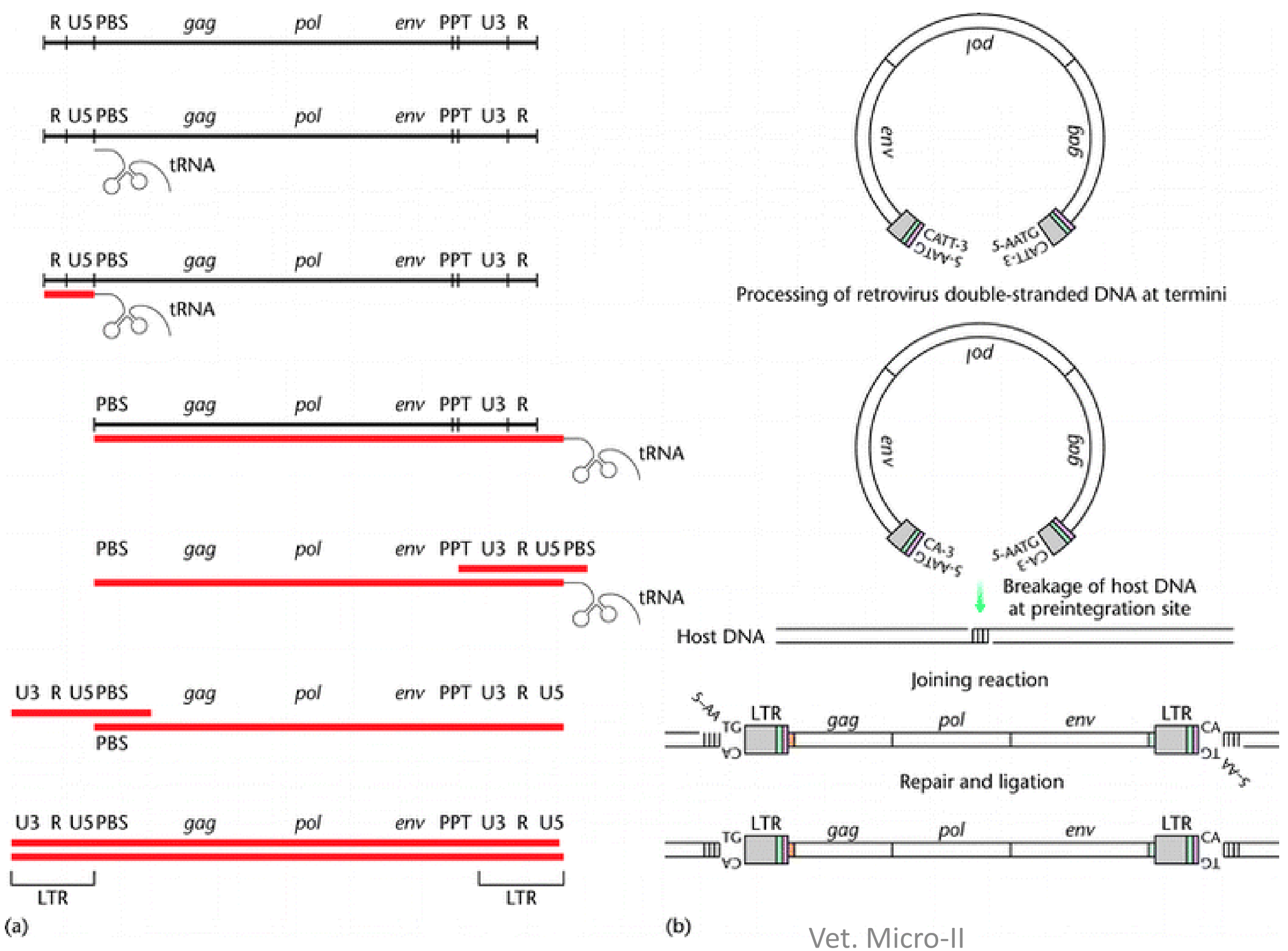



**Figure.** Reverse transcription and integration processes. (a) Reverse transcription. Outline of the reverse transcriptase (RT)-catalysed steps leading from single-stranded genomic RNA (top; black line) to double-stranded proviral DNA (bottom; red line). (b) Integration. The viral DNA (top) is the product of the completed reverse transcription process of (a). Shown are the integrase (IN)-mediated cleavage and religation steps leading to joining of proviral and host DNA. Subsequent repair and ligation are carried out by host factors. Note the loss of two terminal nucleotides of the viral DNA and the generation of a short repeat of host sequences of the integration site.

## Cont…

- Retroviruses can be categorized as exogenous or endogenous.
- Exogenous retroviruses are capable of horizontal transmission between members of the host species.
- Endogenous retroviruses occur widely among vertebrates, constituting up to 10% of the genomic DNA of the host.
  - They are consistently present in germline cells and are transmitted (inherited in Mendelian fashion) only as provirus in germ-cell DNA from parent to offspring.
  - They are regulated by cellular genes and are usually silent but may recombine with exogenous retroviruses.
- Retroviruses in the genera *Alpharetrovirus*, *Betaretrovirus*, *Gammaretrovirus* and *Deltaretrovirus* are frequently referred to as oncogenic retroviruses because they can induce neoplastic transformation in cells which they infect (Table below).
- On the basis of the interval between exposure to the virus and tumour development, exogenous oncogenic retroviruses are designated either as slowly transforming (*cis*-activating) viruses or as rapidly transforming (transducing) viruses.
- Slowly transforming retroviruses induce B-cell, T-cell or myeloid tumours after long incubation periods.
- For malignant transformation to occur, the provirus must be integrated into the host cell DNA close to a cellular oncogene (c-*onc*, protooncogene), resulting in interference with the regulation of cell division (insertional mutagenesis).
- Multiple insertions of the provirus into the host cell genome results in exaggerated gene expression and over-production of a transformation-associated protein.

**Table 63.1** Oncogenic retroviruses of veterinary importance.

| Genus | Virus | Hosts | Comments |
|---|---|---|---|
| *Alpharetrovirus* | Avian leukosis virus | Chickens, pheasants, partridge, quail | Endemic in commercial flocks. Exogenous and endogenous transmission of virus can occur. Causes lymphoid leukosis in birds between 5 and 9 months of age |
| *Betaretrovirus* | Jaagsiekte sheep retrovirus | Sheep | Causes jaagsiekte, a slowly progressive neoplastic lung disease of adult sheep which is invariably fatal. Occurs worldwide except in Australasia |
| | Enzootic nasal tumour virus | Sheep, goats | Closely related to jaagsiekte sheep retrovirus. Causes adenocarcinoma of low-grade malignancy, which affects the nares |
| *Gammaretrovirus* | Feline leukaemia virus | Cats | Important cause of chronic illness and death in young adult cats. Causes immunosuppression, enteritis, reproductive failure, anaemia and neoplasia. Worldwide distribution |
| | Reticuloendotheliosis virus | Turkeys, ducks, chickens, quail, pheasants | Infection usually subclinical. Sporadic disease may present with anaemia, feathering defects, impaired growth or neoplasia. Disease outbreaks have occurred following use of vaccine contaminated with reticuloendotheliosis virus |
| *Deltaretrovirus* | Bovine leukaemia virus | Cattle | Causes enzootic bovine leukosis in adult cattle. A small percentage of infected cattle develop lymphosarcoma |

# Diagnostic approach (General)

- **Sample**: blood, saliva
- **P.M findings**. Histopathology to determine type of tumur
- **Differential diagnosis**: Marek's disease (based on age of affected birds, presence of bursed tumour, absence of thickened peripheral nerves, histology of neoplastic cell types (avian leukosis)
- **Virus isolation** is difficult and not attempted
- Commercial **ELISA** kit for detection group-specific antigen (of ALV )
- Commercial ELISA for antigen detection (major capsid protein p27) (of FLV)
- Immunoflourescent antibody, last for detection of viral antigen in the cytoplasm of leukocytes in blood smears (IFT is more sensitive and more specific than ELISA) and it is a confirmatory test (for FLV)
- **PCR** for detection of the viral nucleic acid

# The End

# Thank you!